

FBXO24 modulates mRNA alternative splicing and MIWI degradation and is required for normal sperm formation and piRNA production

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Abstract

Spermiogenesis is a critical, post-meiotic phase of male gametogenesis, in which the proper gene expression is essential for sperm maturation. However, the underlying molecular mechanism that controls mRNA expression in the round spermatids remains elusive. Here, we identify that FBXO24, an orphan F-box protein, is highly expressed in the testis of humans and mice and interacts with the splicing factors (SRSF2, SRSF3, and SRSF9) to modulate the gene alternative splicing in the round spermatids. Genetic mutation of FBXO24 in mice causes many abnormal splicing events in round spermatids, thus affecting a large number of critical genes related to sperm formation that were dysregulated. Further molecular and phenotypical analyses revealed that FBXO24 deficiency results in aberrant histone retention, incomplete axonemes, oversized chromatoid body (CB), and abnormal mitochondrial coiling along sperm flagella, ultimately leading to male sterility. In addition, we discovered that FBXO24 interacts with MIWI and SCF subunits and mediates the degradation of MIWI via K48-linked polyubiquitination. Furthermore, we show that FBXO24 depletion could lead to aberrant piRNA production in testes, which suggests FBXO24 is required for normal piRNA counts. Collectively, these data demonstrate that FBXO24 is essential for sperm formation and piRNA production by regulating mRNA alternative splicing and MIWI degradation during spermiogenesis.

eLife assessment

This **important** study provides insights into the role of FBXO24 in controlling spermiogenesis and male fertility in mice. The mouse models used and the data are **convincing**. This paper will interest biomedical researchers working on reproductive biology and fertility control.

Introduction

Spermatogenesis is a dynamic but well-organized process in which germ cells develop in the seminiferous tubules of the testis. Spermatogenesis comprises three phases: mitosis of the spermatogonia, meiosis of spermatocytes, and differentiation of round spermatids into mature sperm. The third phase is also known as spermiogenesis. During spermiogenesis, round spermatids undergo many significant changes, including the loss of cytoplasm, migration of cytoplasmic organelles, formation of the acrosome and flagellum, condensation of nuclei, and reorganization of mitochondria around the sperm midpiece (1). In mice, spermiogenesis is subdivided into 16 steps based on changes in acrosome structure and nuclear compaction (2,3). Coordinated regulation of gene expressions in the round spermatids is essential for sperm formation. However, the underlying molecular mechanism that controls mRNA expression during spermiogenesis remains elusive.

FBXO24 (F-Box Protein 24) is a member of the F-box protein family characterized by an F-box domain. The F-box proteins constitute one of the subunits of the ubiquitin protein ligase complex called SCF (SKP1-cullin-F-box). The interaction between FBXO24 and the SCF subunits is unknown. The F-box domain recruits the F-box protein to the SCF complex, and the carboxy-terminal domain is a putative protein-protein interaction domain. According to the different substrate recognition domains, F-box protein was generally divided into three subclasses: F-box/WD repeat-containing protein (FBXW), leucine-rich repeat protein (FBXL), and F-box only protein (FBXO) with/without other unknown domain (4). Compared with the studies of FBXW and FBXL, the roles of most members of the FBXO subfamily have yet to be defined. FBXO24 belongs to the FBXO class. Previous studies demonstrated some FBXO proteins could regulate spermatogenesis. For example, FBXO47 has been reported to be essential for meiosis I progression in the spermatocytes of mouse testis (5,6), and FBXO43 was associated with male infertility and teratozoospermia (7). In addition, FBXO24 has been reported to recognize deacetylated nucleoside diphosphate kinase A (NDPK-A) to enhance its degradation in HeLa cells (8). In H1299 cells, FBXO24 could promote cell proliferation by mediating ubiquitination and degradation of protein arginine methyltransferase 6 (PRMT6) (9). However, the molecular and biological functions of FBXO24 remain unclear, especially its role in male fertility and sperm formation.

In this study, we find that FBXO24 transcript is abundantly expressed in the testis of humans and mice. The loss function of FBXO24 in mice resulted in male sterility, with sperm possessing a collapsed mitochondrial sheath and uncompacted chromatin. We revealed that FBXO24 interacts with splicing factors (e.g., SRSF2, SRSF3, and SRSF9) to mediate gene expressions during spermiogenesis. Unexpectedly, we discovered that FBXO24 mediates MIWI ubiquitination and regulates the chromatoid body (CB) architecture and piRNA production. Together, our study demonstrated that FBXO24 plays a critical role in controlling gene expression in haploid spermatids and sperm formation during spermiogenesis.

Results

FBXO24 expressed in haploid spermatids during spermiogenesis

Multi-alignment and phylogenetic analysis revealed that *Fbxo24* encodes a highly conserved protein expressed in mammals, including humans and mice (Figure S1A). FBXO24 has two domains, F-box and regulator of chromatin condensation 1 (RCC1), which were also highly conserved in humans and mice (Figure S1B). Interestingly, qPCR analysis of multiple mouse organs showed that FBXO24 mRNA was predominantly expressed in mouse testes (Figure 1A). Consistent with this result, transcription expression of analysis of FBXO24 in various human organs from the GE-mini database revealed FBXO24 also a testis-enriched expression in human

testes (**Figure S1C**). Further qPCR analysis of the testis at different developmental stages and purified testicular cells showed that FBXO24 mRNA was highly expressed in the round spermatids and elongating spermatids (**Figure 1B-C**). We used the CRISPR/Cas9-mediated genome-editing system to generate a mouse model containing the *Fbxo24* gene with a C-terminal HA epitope tag (**Figure S2A**) (10). To examine the protein level of FBXO24 in testes, we then characterized the tissue expression pattern of FBXO24 in various organs from adult *Fbxo24*-HA-tagged transgenic mice. Consistent with the mRNA expression showing a testis-enriched expression of FBXO24, FBXO24^{HA-Tag} protein was exclusively expressed in the testes (**Figure 1D**), suggesting FBXO24 plays a role in sperm formation. To explore the subcellular localization of FBXO24 in male germ cells, we performed immunofluorescence analysis by co-staining of FBXO24^{HA-Tag} protein with γ-H2AX (a marker of meiotic DNA damage response) or PNA (an acrosome marker) in adult *Fbxo24*-HA-tagged transgenic mouse testes. The results showed that FBXO24^{HA-Tag} protein was exclusively localized in the nuclei and cytoplasm of spermatids at stages VI-IX (**Figure 1E-F**), demonstrating FBXO24 expressed in the round and elongating spermatids at steps 6-9. These results indicate that FBXO24 is an evolutionarily conserved testis-enriched protein specifically expressed in haploid spermatids during spermiogenesis.

FBXO24 deficiency results in defective spermiogenesis and male infertility in mice

To reveal the role of FBXO24 in male germ cell development and sperm formation, we generated *Fbxo24* knockout (KO) mice using CRISPR/Cas9 gene editing technology. Two small guide RNAs (sgRNAs) were designed to target exon 3, which encodes the conserved F-box domain (**Figure S2B**). To validate nucleotide deletion at the targeted loci, the corresponding genomic regions of *Fbxo24* were evaluated by PCR (**Figure S2C**). qPCR analysis verified the mRNA expression of *Fbxo24* was almost undetectable in *Fbxo24* KO mouse testes (**Figure 2A**). To test the fecundity of *Fbxo24* KO mice, we crossed *Fbxo24* KO mice with fertility-proved WT mice for at least 6 months. The fertility test results showed that *Fbxo24* KO males were completely infertile, but the *Fbxo24* KO females were fertile (**Figure 2B**). Interestingly, we observed the testis size and the ratio of testis weight to body weight were comparable between *Fbxo24* KO males and controls (**Figure 2C** and **Figure S2D**). Consistent with these observations, histological analysis of adult testes and epididymis showed no obvious abnormalities in *Fbxo24* KO mice compared with WT (**Figure S2E**). However, in *Fbxo24* KO mice, we found fewer elongating spermatids (at stages IX-X) and condensing spermatids (at stages XII) and a reduced number of condensed spermatids (at stages II-VIII) (**Figure 2D-E**), concomitant with increased apoptotic signals detected in late spermatids (**Figure S2F-G**), which suggesting FBXO24 deficiency could affect spermatid development during spermiogenesis. To further demonstrate the critical function of FBXO24 during spermiogenesis, we attempted to use the *Fbxo24*-HA-tagged transgenic mice to rescue sperm formation and fertility in *Fbxo24* KO males. As expected, we found that expression of the transgenic FBXO24 could rescue the defects in spermiogenesis and infertility observed in *Fbxo24* KO male mice (**Figure 2F** and **Figure S2H**), which confirmed the specific targeting of *Fbxo24* in the mutants and the true role of FBXO24 in spermiogenesis. Together, these data indicate that depletion of FBXO24 causes defective spermiogenesis, leading to male infertility.

Ablation of FBXO24 in mice causes disorganized mitochondrial sheath of spermatozoa

To elucidate the causes of sterility in *Fbxo24* KO males, we examined sperm count, sperm motility, and sperm morphology in *Fbxo24* KO mice. The results showed that the sperm count and motility were significantly reduced in *Fbxo24* KO mice compared with WT (**Figure 3A-B**). Inspiringly, we found that most of the spermatozoa in *Fbxo24* KO mice displayed a disorganized mitochondrial sheath and had abnormal bending of the flagella (**Figure 3C**). In WT spermatozoa, the midpiece connected smoothly to the principal piece; however, in *Fbxo24* KO spermatozoa, a gap in the

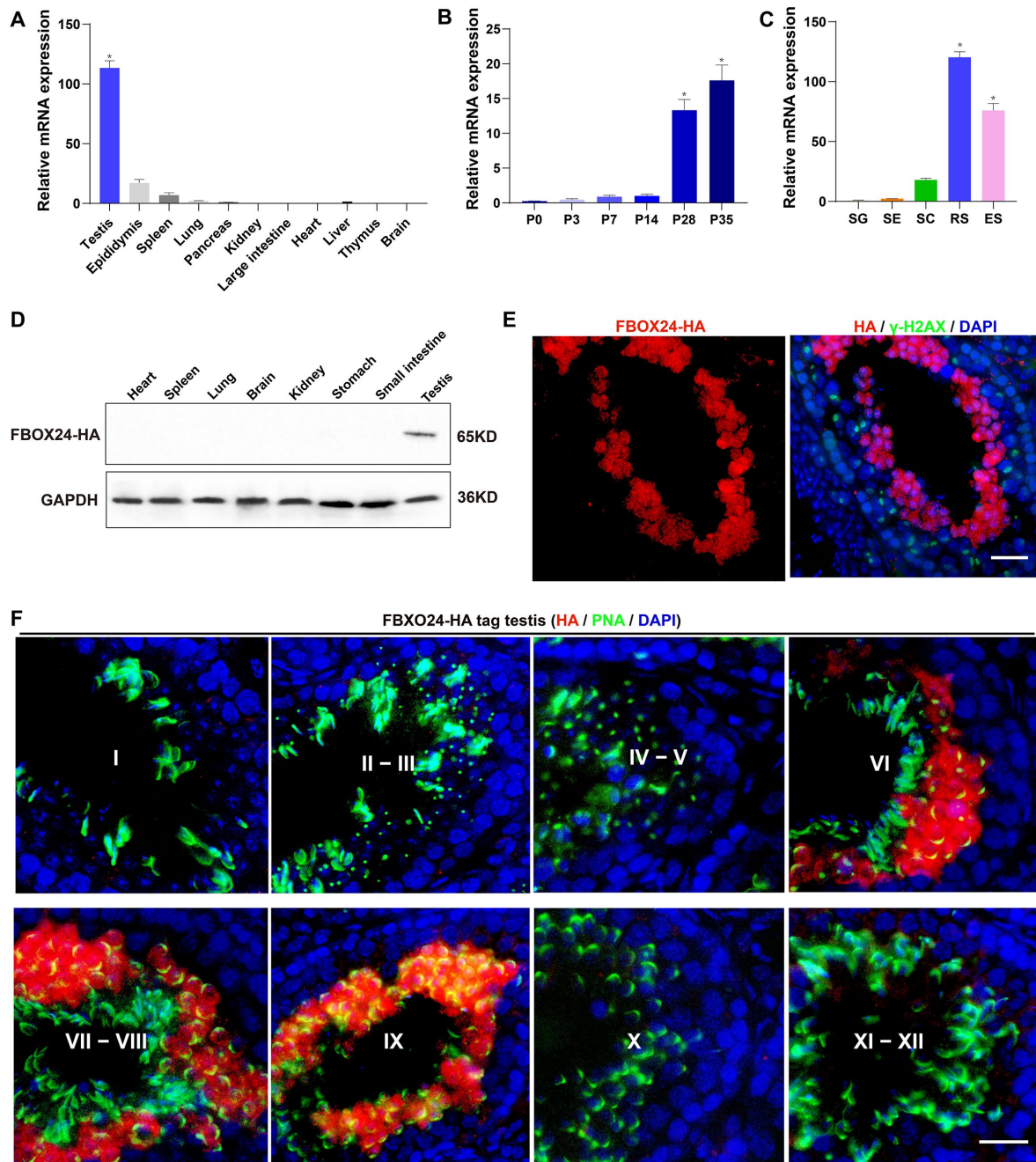


Figure 1.

Expression profiles of FBXO24 during testicular development and spermatogenesis in mice.

(A) qPCR analysis of FBXO24 mRNA levels in multiple organs in mice. $n = 3/\text{group}$. Data are mean $\pm$ S.D. *, $p < 0.05$. (B) qPCR analysis of FBXO24 mRNA levels in developing testes at postnatal day 0 (P0), P3, P7, P14, P28, and P35. $n = 3/\text{group}$. Data are mean $\pm$ S.D. *, $p < 0.05$. (C) qPCR analysis of FBXO24 mRNA levels in isolated spermatogenic cell populations, including spermatogonia (SG), Sertoli cells (SE), spermatocytes (SC), round spermatids (RS), and elongating spermatids (ES). $n = 3/\text{group}$. Data are mean $\pm$ S.D. *, $p < 0.05$. (D) Western blot analysis of FBXO24-HA expression in the tissues of adult transgenic mice. (E-F) Co-immunostaining of FBXO24-HA (red) and (E) γ H2AX (green) or (F) PNA (green) in adult *Fbxo24*-HA testes. Scale bar, 25 μm . DNA (blue) is stained by DAPI. Spermatogenic stages are noted.

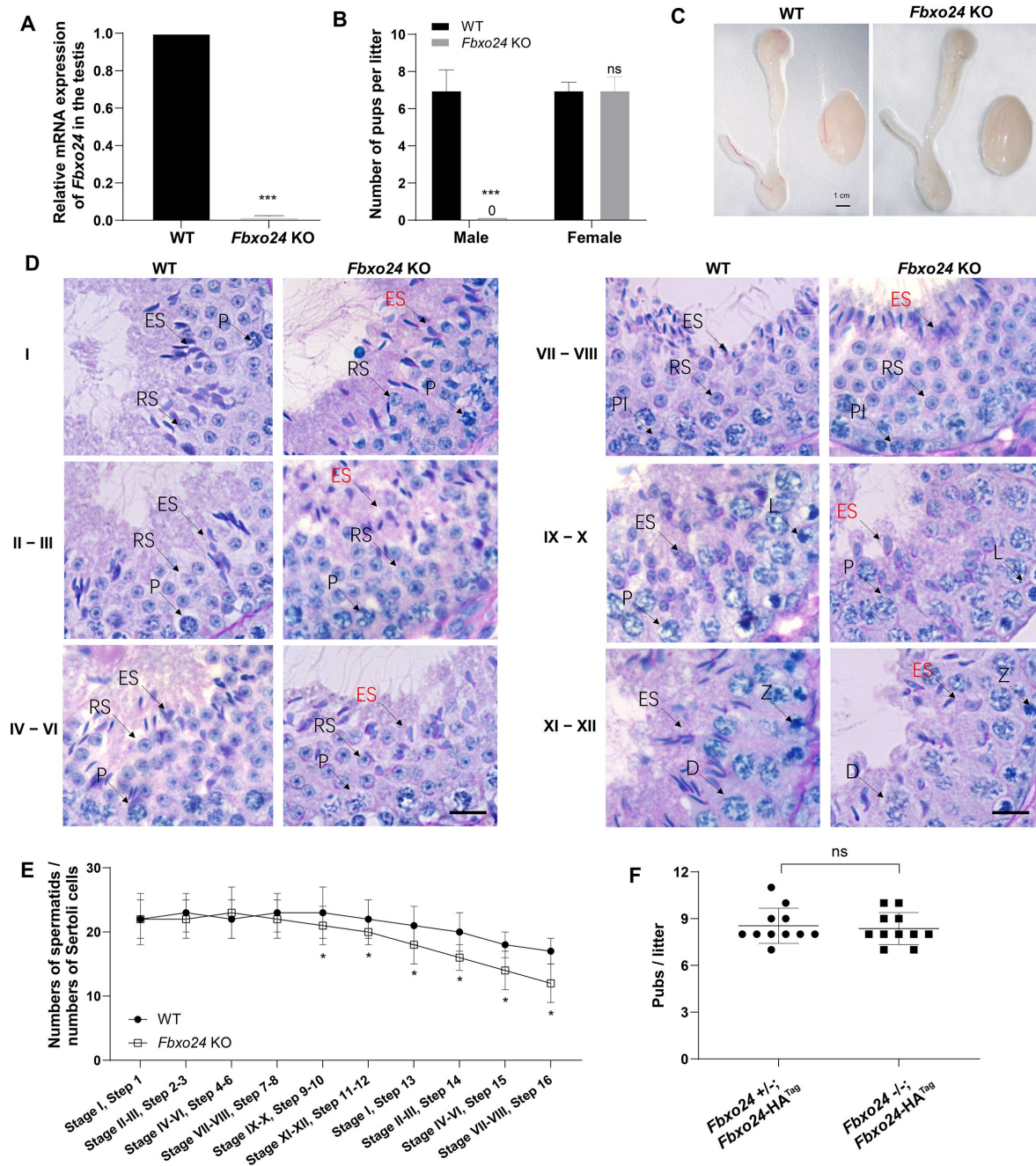


Figure 2.

FBXO24 deletion impairs spermatogenic defects in the late steps of spermiogenesis.

(A) qPCR analysis indicates that FBXO24 mRNA is markedly decreased in the testis of *Fbxo24* knockout (KO) as compared to wild-type (WT). ***, $p < 0.001$. (B) The fertility tests of WT and *Fbxo24* KO male mice mated with fertile female mice are shown. $n = 3/\text{group}$. Error bars represent mean $\pm$ S.D. ***, $p < 0.001$; ns: not significant. (C) Histological images of testes and epididymis of WT and *Fbxo24* KO mice at 8-weeks old are shown. (D) PAS-hematoxylin staining of *Fbxo24* KO testis at 8-weeks old contained less-condensed late spermatids (red arrows). Spermatogenic stages are noted. RS, round spermatids; ES, elongating spermatids; PI, preleptotene; P, pachytene; Z, zygotene; D, diplotene. Scale bars = 25 μm . (E) The number of late spermatids is significantly reduced in *Fbxo24* KO testis. Ratios of spermatids and Sertoli cells in tubule cross-sections of specific stages of seminiferous epithelial cycles and corresponding spermatid development steps are shown. Data are mean $\pm$ S.D. *, $p < 0.05$. (F) Litter sizes of mating tests. F1 generation of intercrosses between the indicated males and *Fbxo24*^{+/-} females are shown. Each dot represents one litter.

midpiece can be observed (**Figure 3C-D** [↗](#)), leading to many sperm with an abrupt bending of the tail in the region of the neck and midpiece (**Figure S3A**). In addition, we observed the midpiece length was significantly shorter in *Fbxo24* KO mouse spermatozoa compared with WT ($21.67 \pm 1.524 \mu\text{m}$ in WT vs. $14.80 \pm 3.430 \mu\text{m}$ in *Fbxo24* KO) (**Figure 3E** [↗](#)). To better reveal spermatozoon abnormalities in *Fbxo24* KO mice, we decided to examine the ultrastructure of the spermatozoa using a scanning electron microscope (SEM). In WT spermatozoa, the midpiece was normally populated with many mitochondria in a spiraling fashion. However, in *Fbxo24* KO spermatozoa, mitochondria near the annulus were absent, resulting in a shorter mitochondrial sheath (**Figure 3F** [↗](#)). Transmission electron microscope (TEM) further confirmed that mitochondria were not present near the annulus or neck and left apparent gaps along the mitochondrial sheath in *Fbxo24* KO spermatozoa (**Figure 3G** [↗](#)), indicating mitochondrial sheath formation was disrupted. In addition, some mitochondria in the midpiece of *Fbxo24* KO spermatozoa appeared large and vacuolar. To determine whether the failure in the assembly of the mitochondrial sheath could cause disrupted flagellar development, we examined cross-sections of the *Fbxo24* KO sperm flagellum. In the midpiece, well-defined outer dense fibers (ODFs) and the axoneme consisting of the typical “9 + 2” microtubules were observed in WT spermatozoa, whereas the *Fbxo24* KO spermatozoa displayed incomplete axonemes with missing axonemal microtubules (**Figure 3H** [↗](#)). In the principal piece, the *Fbxo24* KO spermatozoa also showed completely or partially lacking ODFs and a typical “9 + 2” microtubular structure (**Figure 3H** [↗](#)). These severe flagellar defects may explain why *Fbxo24* KO spermatozoa displayed decreased motility (**Figure 3B** [↗](#)).

We next asked whether FBXO24 deletion influences the protein levels of mitochondrial and flagellum components in spermatozoa through Western blot analysis of spermatozoa. We found that the levels of MFN2 (a marker of mitochondrial fusion) and DRP1 (a marker of mitochondrial fission) were decreased in *Fbxo24* KO spermatozoa (**Figure 3I** [↗](#) and **Figure S3B**). Moreover, the protein expression of ODF2, AKAP3, and TSSK4, which were directly related to the formation of the sperm tail ([11](#) [↗](#)–[13](#) [↗](#)), were all decreased in *Fbxo24* KO spermatozoa (**Figure 3J** [↗](#) and **Figure S3B**). This was supported by immunofluorescence analysis of ODF2 and AKAP3 with a reduced signals in *Fbxo24* KO flagella (**Figure S3C-D**). Interestingly, we found that the level of the CUL1, a scaffold for E3 ubiquitin ligase, was also significantly reduced in *Fbxo24* KO spermatozoa (**Figure 3J** [↗](#) and **Figure S3B**). Together, these results indicate that FBXO24 is critical for mitochondrial sheath formation and could modulate the protein expression levels of mitochondrial and flagellar components to maintain sperm motility.

The mitochondrial and chromatoid body architecture were affected in FBXO24-deficient round spermatids

To further explore whether the mitochondrial organelle morphology in round spermatids was affected upon FBXO24 depletion, we examined the ultrastructure of WT and *Fbxo24* KO testes by TEM. Notably, the mitochondria in the round spermatids of *Fbxo24* KO testis displayed vacuolar and had abnormal central cristae (**Figure 4A-B** [↗](#)). Interestingly, we identified that the chromatoid body (CB) was structurally maintained in *Fbxo24* KO round spermatids, whereas the size was often larger than WT (**Figure 4C-D** [↗](#)). CB was considered to have the functions in the assembly of the mitochondria during the spermiogenesis, and dysfunctional transformation of the CB in mouse spermatids could cause spermatozoa to possess a collapsed mitochondrial sheath ([14](#) [↗](#)), which resembled the phenotype observed in our *Fbxo24* KO mice. To explore the molecular reason for the abnormal CB formation, we examined the protein expression levels of CB components in WT and *Fbxo24* KO testes. Since MIWI was a core component of CB and also identified as an FBXO24 interacting partner from our immunoprecipitation-mass spectrometry (IP-MS) (**Table S1**), we focused on the examination of MIWI expression between WT and *Fbxo24* KO testes. Western blot analyses revealed that the protein level of MIWI was remarkably increased in the *Fbxo24* KO testes compared with WT (**Figure 4E** [↗](#)). However, the level of other proteins of CB, and PRMT6

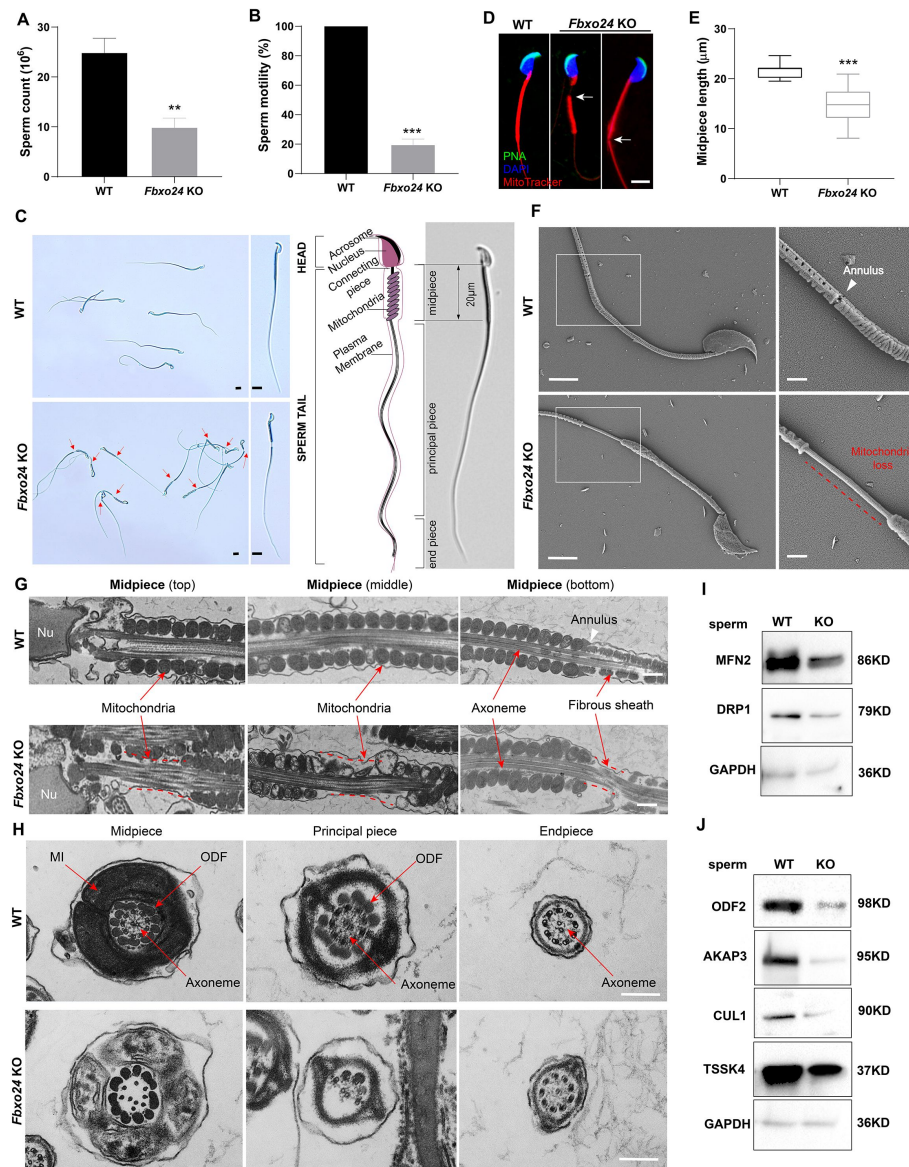


Figure 3.

Sperm mitochondria and flagella are defective in FBXO24-deficient mice.

(A-B) Quantification of sperm counts (A) and sperm motility (B) from WT and *Fbxo24* KO epididymis are shown. n = 3/group. Error bars represent mean $\pm$ S.D. **, $p < 0.01$. ***, $p < 0.001$. (C) Sperm morphological images show the defective sperm of *Fbxo24* KO mice. Red arrows indicate abnormal gaps in the mitochondrial sheath. Scale bars = 5 μ m. (D) Immunofluorescence images of sperm from WT and *Fbxo24* KO epididymis. PNA (acrosome, green), MitoTracker (mitochondria, red), and DAPI (nucleus, blue). White arrows indicate the weak or absent staining of MitoTracker. Scale bars = 5 μ m. (E) Quantifications of the length of sperm midpiece from WT and *Fbxo24* KO mice are shown. n = 100/group. Error bars represent mean $\pm$ S.D. ***, $p < 0.001$. (F) Scanning electronic microscopy (SEM) images indicate the mitochondria detachment from the flagellum of *Fbxo24* KO sperm. Right panel insets show higher magnification of sperm midpiece. The arrowheads indicate the annulus. The dashed red line indicates a region where mitochondria are absent. Scale bars = 2 μ m. (G) Transmission electronic microscopy (TEM) images indicate the mitochondria defects in three regions of *Fbxo24* KO sperm midpiece in the longitudinal sections. Nu, Nucleus. The dashed red lines indicate a region where mitochondria are absent. The white arrowhead indicates sperm annulus. Scale bar = 0.2 μ m. (H) TEM images indicate the ultrastructure of the midpiece, principal piece and end piece of WT and *Fbxo24* KO sperm flagellum in the cross-sections. Arrows indicate the mitochondria (MI), outer dense fiber (ODF) and axoneme. Scale bar = 0.2 μ m. (I-J) Western blot shows the levels of proteins of mitochondria (I) and axoneme (J) of WT and *Fbxo24* KO sperm. GAPDH serves as a loading control.

(reported to be the substrate of FBXO24)(9) appeared to be unchanged in *Fbxo24* KO testes compared with WT testis (Figure 4E). These results suggest that FBXO24 is essential for maintaining the normal architecture of mitochondria and CB in the round spermatids.

Histones failed to be replaced in FBXO24-deficient mouse spermatozoa

To investigate the nuclear morphology of spermatozoa, we examined the sperm head structures of WT and *Fbxo24* KO spermatozoa by SEM. The *Fbxo24* KO spermatozoa exhibited abnormal head morphologies that deviated from the flat, crescent-shaped structures of their WT counterparts (Figure 5A). Specifically, the defects of *Fbxo24* KO spermatozoa included a disorganized anterior acrosome (AS) and the absence of a distinct equatorial segment (EQ), post-acrosomal sheath (PAS), ventral spur (VS), and sharp hook rim (HR) (Figure 5A). TEM analysis further revealed less condensed nuclei of *Fbxo24* KO spermatozoa (Figure 5B), suggesting an abnormal nuclear chromatin compaction was affected in *Fbxo24* KO sperm head. *Fbxo24* KO spermatozoa exhibited elevated levels of DNA damage by TUNEL analysis (Figure S3E-F). In many knockout mice studies, impaired chromatin condensation is frequently associated with abnormal sperm head morphology (15). We then investigated the potential causes of abnormal head structures in *Fbxo24* KO spermatozoa. Given that impaired histone-to-protamine exchange can result in less-condensed nuclei in spermatozoa, we examined histone and protamine levels in *Fbxo24* KO mouse sperm. Western blot revealed increased levels of all core histones H2A, H2B, H3, H4, and transition protein TNP1 but reduced levels of PRM2 in *Fbxo24* KO sperm (Figure 5C-D). We found that FBXO24 did not have the interactions with histones H2A, H2B, H3, H4 and transition protein TNP1 (Figure S3G). Moreover, we found a reduced protein level of PHF7, TSSK6, and RNF8 in *Fbxo24* KO testis (Figure 5E-F), which have been reported to regulate the chromatin structure to facilitate histone-to-protamine replacement (1,16,17). These data suggest that FBXO24 deletion causes an incomplete histone-to-protamine exchange and defective chromatin compaction during spermiogenesis.

FBXO24 depletion leads to aberrant expression of spermiogenic genes in the round spermatids

To further investigate the underlying molecular changes in sperm formation upon FBXO24 depletion, we performed RNA-seq using purified round spermatids (RS) from adult *Fbxo24* KO and WT males (Figure S4A). RNA-seq analysis showed a large number of genes significantly differentially expressed in *Fbxo24* KO round spermatids, including 5368 upregulated and 4097 downregulated genes (Figure 6A and Table S2). The Gene Ontology (GO) analysis revealed that the downregulated genes were related to mitochondrion localization, chromatin organization, and protein K48-linked ubiquitination (Figure 6B and Table S3). The expression of many critical genes involved in mitochondrion localization (e.g., *Hif1a*, *Nefl*, *Mgarp*, *Zmynd12*, *Map7d1*, *Bbs7*, *Mfn1*, *Ube2j2*) and chromatin organization (e.g., *Rnf20*, *Ep300*, *Sirt1*, *Hmgb2*, *Hsf1*, *Cul4b*, *Kat5*, *Phf7*) were significantly decreased in *Fbxo24* KO round spermatids (Figure 6C-D). Interestingly, we found that *Sept4* and *Sept12*, the members of *Septin* family known to generate annuli of spermatozoa, were also downregulated in *Fbxo24* KO round spermatids (Figure S4B). The most upregulated genes in *Fbxo24* KO round spermatids were mainly related to cellular metabolic processes and organelle organization (Figure S4C). Furthermore, we identified an overlap between the downregulated genes and the altered alternative splicing (AS) genes in *Fbxo24* KO round spermatids (Figure 6E). A large number of mRNA splicing events were identified in *Fbxo24* KO round spermatids (Figure 6F), including skipped exons (SE), alternative 5' splice sites (A5SS), alternative 3' splice sites (A3SS), mutually exclusive exons (MXE), and retained introns (RI). Interestingly, GO analysis of the abnormally spliced genes highlighted the functional categories involved in sperm formation, such as outer dynein arm assembly, mitochondrion localization, and chromatin organization, et al. (Figure S4D). Thus, the RNA-seq data revealed

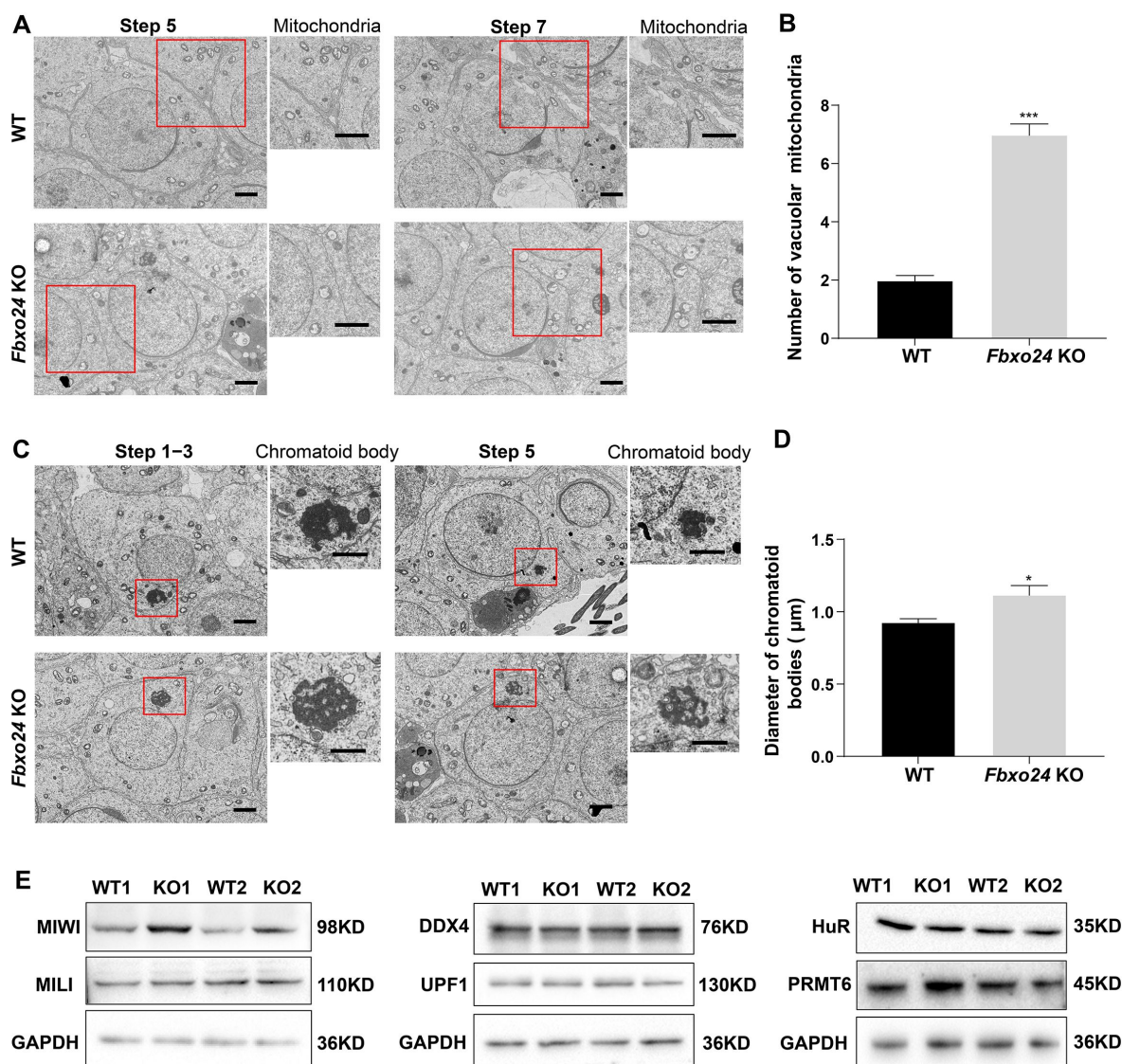


Figure 4.

Ablation of FBXO24 affects mitochondria and chromatoid body architecture in the round spermatids.

(A) TEM images show vacuolar mitochondria with disorganized cristae in the round spermatids of *Fbxo24* KO testes. Right panel insets show higher magnification of mitochondria. Scale bars = 1 μm. (B) Quantification of the number of vacuolar mitochondria. ***, $p < 0.001$. (C) TEM images showing decondensed and enlarged chromatoid body (CB) with an irregular network in the round spermatids of *Fbxo24* KO testes. Right panel insets show a higher magnification of CB. Scale bars = 1 μm. (D) Quantification of size/diameters of CB. *, $p < 0.05$. (E) Western blot analysis expression levels of CB components and PRMT6 in testes from WT and *Fbxo24* KO mice at 8 weeks old. GAPDH serves as a loading control.

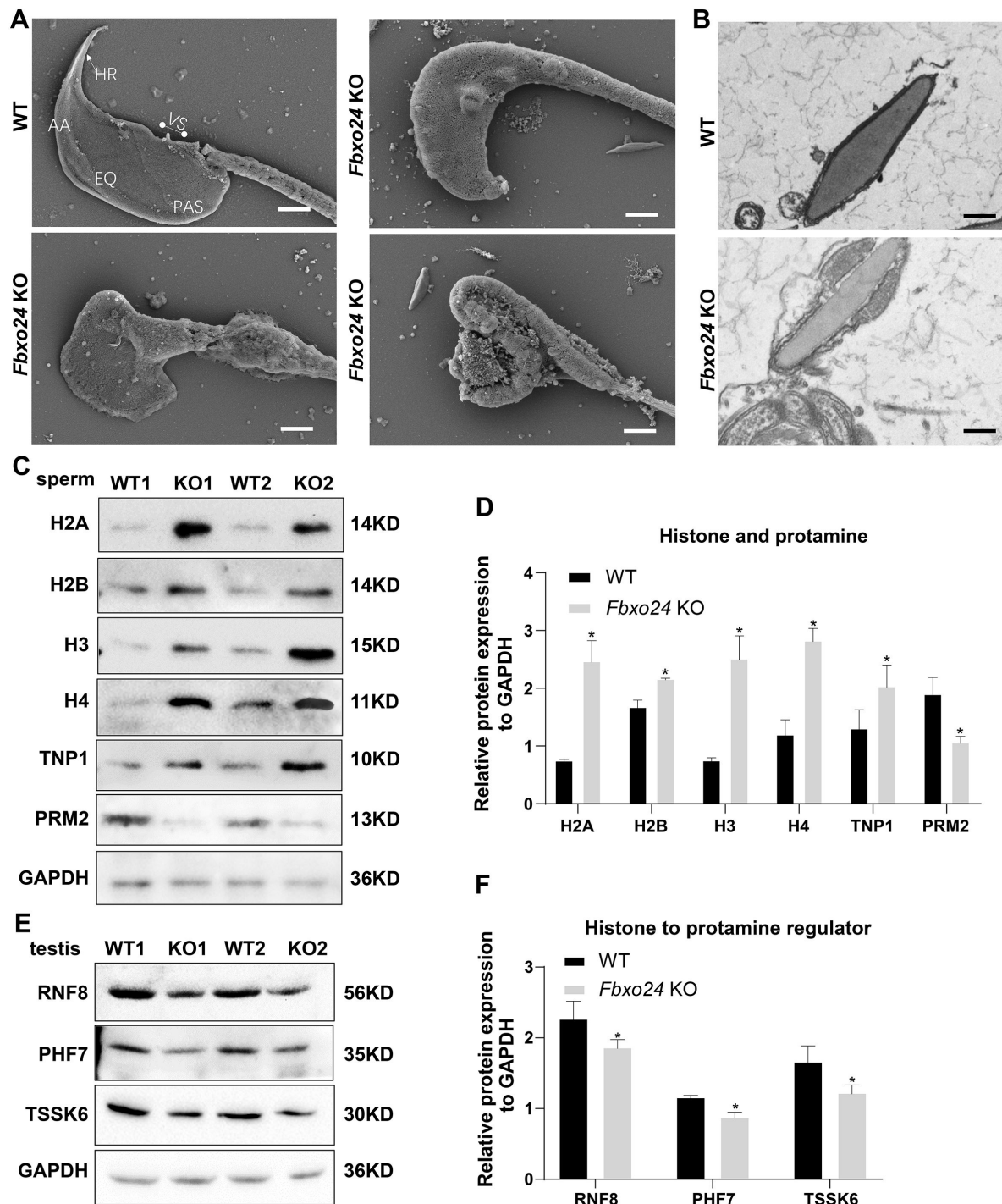


Figure 5.

FBXO24 deficiency in mice impairs sperm histone-to-protamine exchange.

(A) SEM images show the abnormality of *Fbxo24* KO sperm head. AA, anterior acrosome; EQ, equatorial segment; PAS, postacrosomal segment; VS, ventral spur; HR, hook rim. Scale bars = 2 μ m. (B) TEM images show the decondensed nucleus (Nu) of *Fbxo24* KO sperm. Scale bars = 1 μ m. (C) Western blot analysis of the expression of histones (H2A, H2B, H3, and H4), transition proteins (TNP1), and protamines (PRM2) from WT and *Fbxo24* KO sperm are shown. GAPDH serves as a loading control. (D) Quantification of protein levels. Error bars represent mean $\pm$ S.D., $n = 3$. *, $p < 0.05$. (E) Western blot analysis of the expression of indicated proteins in WT and *Fbxo24* KO testis. GAPDH serves as a loading control. (F) Quantification of protein levels. Error bars represent mean $\pm$ S.D., $n = 3$. *, $p < 0.05$.

genetic ablation of *Fbxo24* has a significant impact on the transcriptome of round spermatids, indicating that FBXO24 is required for gene expression in round spermatids during spermiogenesis.

Loss of FBXO24 results in aberrant mRNA alternative splicing in the round spermatids

Since FBXO24 expressed in the nucleus of the round spermatids, we speculated that FBXO24 might interact with the key splicing factors. To test this hypothesis, we performed immunoprecipitation-mass spectrometry using the HA antibody to unbiasedly identify the interactome of FBXO24 in testes. The IP-MS results showed many splicing factors, such as SRSF2, SRSF3, and SRSF9, highly enriched in HA-immunoprecipitates (**Table S1**). Through co-immunoprecipitation (Co-IP) assays, we confirmed FBXO24 has indeed interacted with splicing factors SRSF2, SRSF3, and SRSF9 in the testes (**Figure 7A**). Interestingly, we found a significant decrease in SRSF2, SRSF3, and SRSF9 protein expression levels in *Fbxo24* KO testes compared to controls (**Figure 7B-C**). Because many alternative splicing genes related to sperm formation were affected in *Fbxo24* KO round spermatids by RNA-seq analysis (**Figure 6E**), we asked if the splicing levels of corresponding exons of those genes were altered. To this end, we performed RT-PCR on purified round spermatids to verify the alternative splicing alteration. The RT-PCR results showed that many genes related to sperm formation, including mitochondria (*Mfn1*), flagellum (*Zmynd12*, *Map7d1*, and *Bbs7*), chromatin (*Phf7* and *Kat5*), acrosome (*Nuch2*), appeared to be significantly alternative splicing changes in their corresponding exons (**Figure 7D-G**). In addition, we found that the alternative splicing of *Ube2j2*, a ubiquitin-conjugating enzyme, was also affected in *Fbxo24* KO round spermatids (**Figure 7H**). Of note, sperm motility of *Ube2j1* KO mice was also severely impaired (18). Together, these results indicate that FBXO24 could interplay with key splicing factors to regulate the mRNA splicing of some essential genes related to sperm formation.

FBXO24 mediates K48-linked polyubiquitination of MIWI

Since the increased expression of MIWI was found in the testes of *Fbxo24* KO mice, we sought to determine whether FBXO24 interacts with MIWI and affects its degradation via ubiquitination. We ectopically expressed FBXO24-mCherry in HEK293T cells and incubated FBXO24-mCherry with the testicular protein lysate. MIWI and CUL1 were detected in FBXO24 immunoprecipitates (**Figure 8A**), suggesting that FBXO24 could bind MIWI and CUL1 *in vivo*. To confirm that FBXO24 endogenously interacted with its candidates, we performed immunoprecipitation with the testis of *Fbxo24*-HA-tagged transgenic mice. Through co-immunoprecipitation assays, we identified that FBXO24 indeed interacts with MIWI and SCF subunits, including CUL1, SKP1 and RBX1. (**Figure 8B**). To further dissect the binding abilities between FBXO24 and MIWI, we examined which domain was responsible for the interaction through the truncated proteins of FBXO24, which only contain the F-box or RCC1 domain (**Figure 8C**). The immunoprecipitation results showed that both the F-box and RCC1 domains of FBXO24 could bind with MIWI (**Figure 8D**). Furthermore, we found that FBXO24 leads to a decreased level of MIWI in a concentration-dependent manner, suggesting FBXO24 modulates the degradation of MIWI (**Figure 8E**). Because K48- and/or K63-linked polyubiquitination was reported to be responsible for degradation, trafficking, and phosphatase activation (19), we then asked which kind of modification participates in MIWI ubiquitination. We found that overexpression of FBXO24 enhanced polyubiquitination of MIWI in the presence of Ub(K48). We also used a construct encoding Ub(K63). All lysine residues of it are mutated to R except K63. We found that overexpression of FBXO24 did not enhance polyubiquitination of MIWI in the presence of Ub(K63) (**Figure 8F**). Consistently, FBXO24 contributes to the decreased expression of MIWI in a ubiquitin-dependent manner (**Figure 8G**). The ubiquitin level of MIWI was reduced in FBXO24 KO round spermatids (**Figure 8H**). These data suggest that FBXO24 interacts with SCF subunits and mediates the degradation of MIWI via K48-linked polyubiquitination.

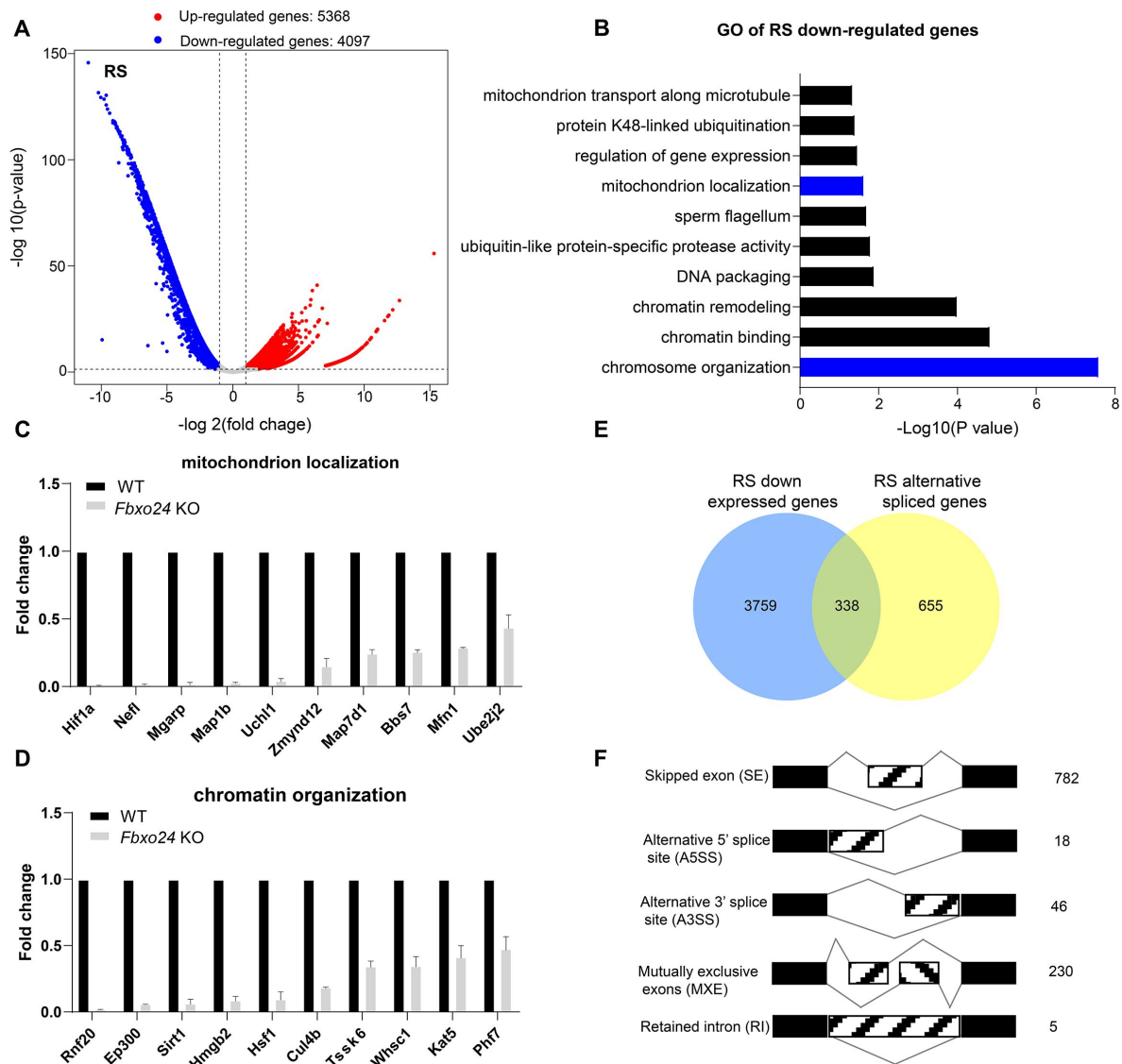


Figure 6.

RNA-seq analyses of the round spermatids from FBXO24-deficient testes.

(A) Volcano plot of differentially expressed transcripts in the round spermatids (RS) of *Fbxo24* KO vs. WT mice. Each red (up-regulation) or blue (down-regulation) dot represents a significantly changed gene. (B) GO term enrichment analysis of downregulated transcripts of *Fbxo24* KO RS. (C-D) Gene expressions of mitochondrion localization (C) and chromatin organization (D) in RNA-seq analysis. (E) Venn diagrams showing the overlap between downregulated genes and abnormal alternative splicing genes in *Fbxo24* KO RS. (F) Summary of differential splicing events in *Fbxo24* KO RS. The number of each category of alternative splicing is indicated.

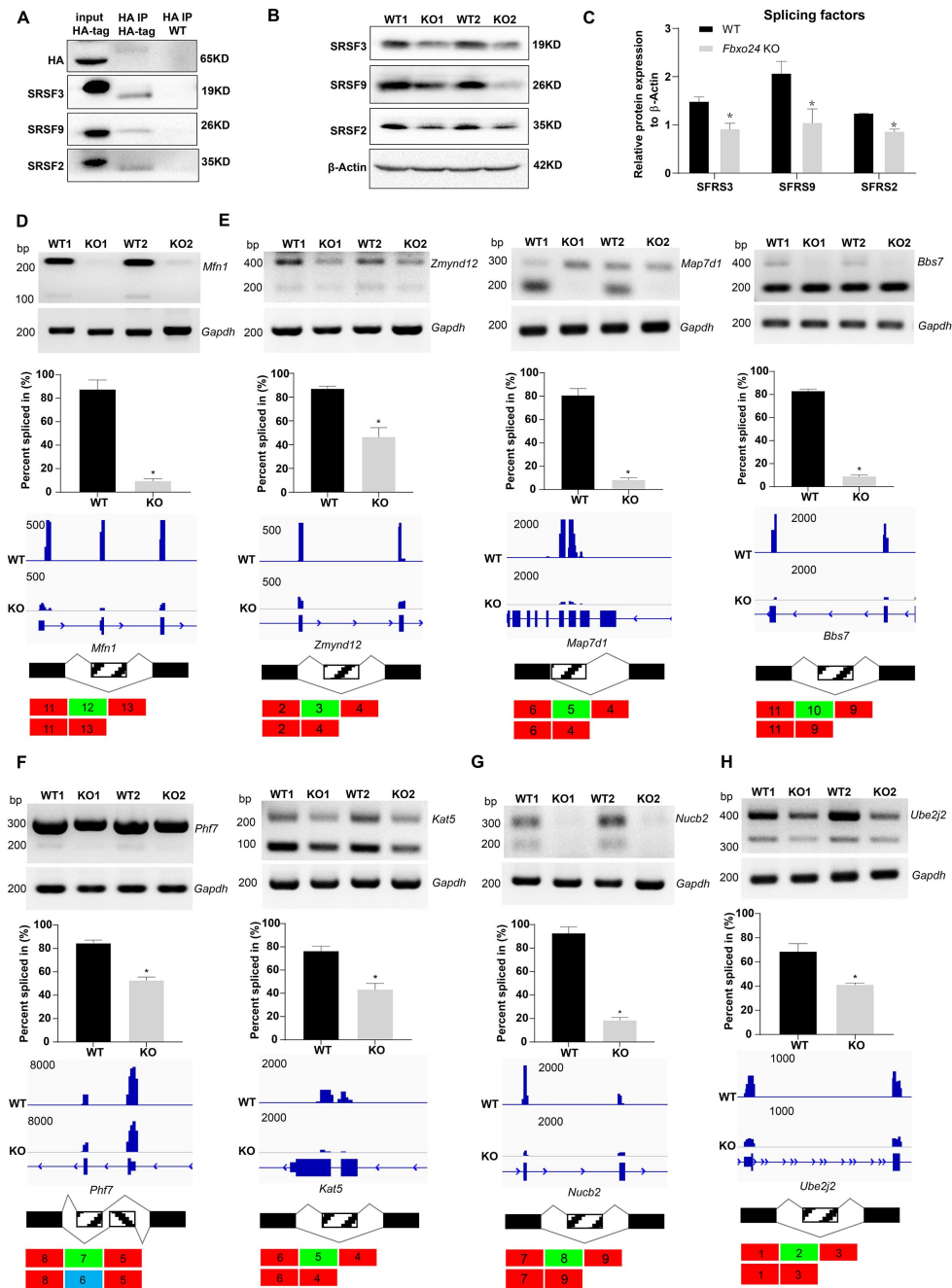


Figure 7.

Aberrant alternative splicing of spermiogenesis genes in the round spermatids of FBXO24-deficient mice.

(A) Co-immunoprecipitation analysis of FBXO24 and the splicing regulators (SRSF2, SRSF3, and SRSF9) in *Fbxo24*-HA tagged mice testis. WT testis was used as a negative control. (B) Western blotting analysis of the splicing regulators in WT and *Fbxo24* KO testis. GAPDH was used as a loading control. (C) Quantification of protein levels. Error bars represent mean $\pm$ S.D., $n = 3$. *, $p < 0.05$. (D-H) Validation of abnormal alternative splicing genes related to (D) mitochondria (*Mfn1*), (E) flagellum (*Zmynd12*, *Map7d1*, and *Bbs7*), (F) chromatin (*Phf7* and *Kat5*), (G) acrosome (*Nucb2*), and (H) ubiquitination (*Ube2j2*). The top panels represent RT-PCR analysis of indicated genes in WT and *Fbxo24* RS. *Gapdh* serves as a loading control. The middle panels show the quantification of percent spliced in (PSI) and alternative sites in RNA-seq. Error bars represent mean $\pm$ S.D., $n = 2$. *, $p < 0.05$. The bottom panels represent the schematic diagram of alternative sites exons.

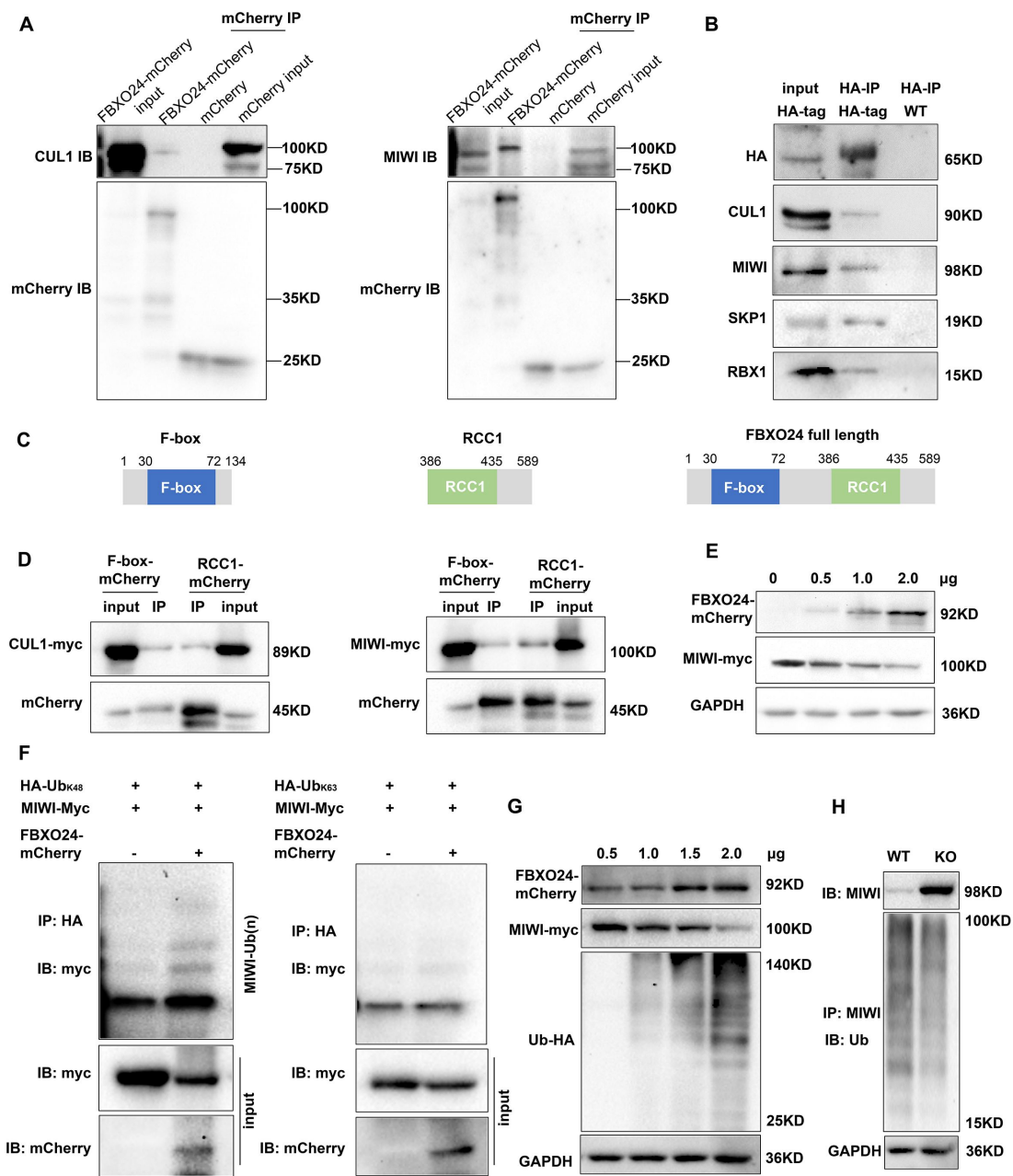


Figure 8.

FBXO24 interacts with MIWI and mediates its K48-linked polyubiquitination.

(A) HEK293T cells transfected with empty FBXO24-mCherry or mCherry vector. Anti-mCherry beads were used for immunoprecipitation (IP), and western blots were used to detect the CUL1 (left panel) and MIWI (right panel) expression. (B) The testis lysate of *Fbxo24*-HA tagged mice were immunoprecipitated with anti-HA beads. Western blots were used to detect the HA, CUL1, MIWI, SKP1, and RBX1 expression. (C) Schematic structures of the truncated FBXO24 protein are shown. Broken boxes show the domain of F-box and regulator of chromosome condensation 1 (RCC1). (D) HEK293T cells were transfected with indicated plasmids. Immunoprecipitation was performed using the anti-mCherry antibody. (E) Western blot analysis of HEK293T cells transfected with indicated FBXO24-mCherry and 2µg MIWI-myc plasmids. The cell lysates were immunoblotted with anti-mCherry and anti-myc antibodies. (F) FBXO24 mediated the ubiquitination of MIWI in the presence of Ub(K48) not Ub(K63). (G) HEK293T cells were transfected with indicated FBXO24-mCherry, 2µg MIWI-myc, and 2µg Ub-HA plasmids. The cell lysates were immunoblotted with the anti-mCherry, anti-myc, and anti-HA antibodies. GAPDH serves as a loading control. (H) Ubiquitination analysis of MIWI in the round spermatids of *Fbxo24* KO mice. The cells were treated with MG132 (10 µM) in the ubiquitination assay.

FBXO24 is required for normal piRNA production in the testes

Because piRNAs could be loaded onto MIWI (20), we next examined whether the population of piRNAs was affected in *Fbxo24* KO testes. Sequencing of small RNA libraries constructed from total RNA revealed that the expression of miRNAs was not extensively changed in *Fbxo24* KO testes, with only 20 up- and 13 down-regulated miRNAs (Figure 9A, Figure S5A, and Table S4). However, a large number of piRNAs were identified to be differentially expressed in *Fbxo24* KO testes, 463 up- and 128 down-regulated piRNAs (Figure S5B and Table S5). After normalized with miRNA counts, the relative amount of total piRNAs was also increased in *Fbxo24* KO testes (Figure S5C). Specifically, an increased piRNAs profile was observed in the genomic regions between 2 kb upstream of transcriptional start sites (TSS) and 2 kb downstream of transcriptional end sites (TES) in *Fbxo24* KO testes (Figure 9B). To assess the potential functions of the association of piRNAs to different functional gene regions, we analyzed the piRNAs mapping density of varying gene regions, including coding region (CDS), 5' and 3' untranslated region (UTR), and intron, as well as transposable element (TE), including retrotransposon (LTR, LINE, and SINE) and DNA transposon. The results showed that the intron- and retrotransposon-derived piRNAs were the most affected types in *Fbxo24* KO testes (Figure 9C and Table S6). Furthermore, we analyzed the abundance and size of piRNA populations in adult *Fbxo24* KO and WT testes. The results showed that the number of 29–31nt piRNAs were remarkably increased in *Fbxo24* KO testes compared with WT (Figure 9D). In addition, we found many up-regulated piRNAs in *Fbxo24* KO testes were also presented in previously published data of MIWI immunoprecipitates (21) (Figure 9E), suggesting that the increased piRNAs were MIWI-bound piRNAs. The first nucleotide and 10th nucleotide of repeat piRNAs of *Fbxo24* KO testes exhibited a similar strong bias compared with piRNAs of WT testes, indicating the amplification process of the ping-pong cycle was not affected (Figure S5D). Together, these results show that the loss function of FBXO24 results in aberrant piRNA production in testes, suggesting FBXO24 is related to normal piRNA counts of the testis.

Discussion

In this study, we identified FBXO24 as a testis-enriched F-box protein conserved in mammals and highly expressed in the round spermatids and elongating spermatids. To define its biological roles *in vivo*, we generated genetically engineered *Fbxo24* knockout and *Fbxo24* HA-tagged transgenic mouse models. Our functional study demonstrated that FBXO24 is essential for spermiogenesis and piRNA production. FBXO24 deficiency leads to remarkable histone retention and the disruption of histone-to-protamine transition in the mature sperm. A growing number of studies have demonstrated impaired chromatin condensation can cause nuclear damages as DNA denaturation or fragmentation often associated with male infertility (22). The molecular basis of nuclear condensation is related to key events such as the expression of testis-specific histone variants, post-translational modifications of histones, and transient DNA strand breaks. Indeed, we found that, in the current study, FBXO24 could modulate the homeostasis of the histone-to-protamine transition regulators PHF7 (17), TSSK6 (1), and RNF8 (16), while RNF8 was reported to be independent to the process (23).

Although the abnormalities of mitochondrial sheaths are observed in infertile men (24), the key proteins implicated in sperm mitochondrial sheath formation largely remain unclear due to a lack of good animal models with the typical phenotype. Notably, we found that the round spermatids and spermatozoon of *Fbxo24* KO mice have impaired mitochondrial architecture, with mitochondrial size heterogeneity and a more open structure of the mitochondria with less crista. The defective mitochondria were associated with energetic disturbances and reduced sperm motility. Using MitoTracker staining and an electron microscope, we demonstrated the mitochondrial structure is severely disrupted in *Fbxo24* KO spermatozoa. Previous studies showed abnormalities of the annulus (*Sept4* and *Sept12*) (25,26), and chromatid bodies (*Tssk1* and *Tssk2*) (14,27) can disrupt abnormal mitochondrial coiling. The chromatid body was

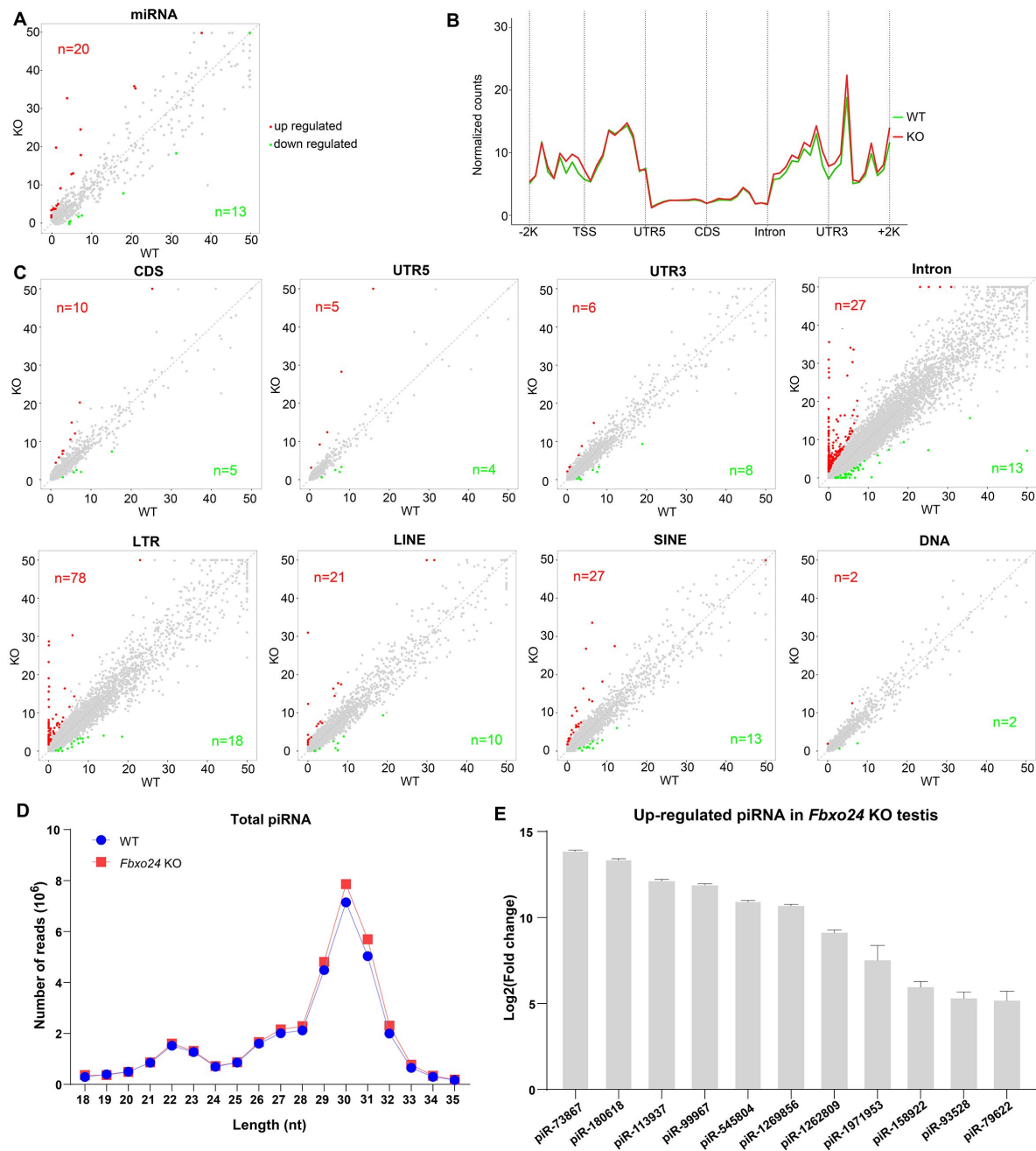


Figure 9.

Small RNA-seq analysis of testes from FBXO24-deficient mice.

(A) A scatter plot of differentially expressed miRNA is shown. Red and green dots represent up- and down-regulated miRNA (fold change > 2, $p < 0.05$), respectively. (B) Genomic distribution of piRNA profile in *Fbxo24* KO vs. WT testis. piRNA levels were examined in each 200 bp interval of a 2 kb region upstream and downstream of the annotated genes. (C) Scatter plots of differentially expressed piRNA mapping density (reads/kb) of the coding region (CDS), 5' and 3' untranslated region (UTR), and intron, as well as transposable element (TE), including retrotransposon (LTR, LINE, and SINE) and DNA transposon. The piRNA read counts were normalized with miRNA. Red and green dots represent up- and down-regulated piRNA (fold change > 2, $p < 0.05$), respectively. (D) The size distribution of piRNAs in *Fbxo24* KO vs. WT testis. (E) The top 10 up-regulated piRNAs in *Fbxo24* KO testis exist in MIWI immunoprecipitates of GSM822760 data.

considered to migrate to the caudal end of the developing middle piece of the flagellum, moving in front of the mitochondria that are engaged in mitochondrial sheath morphogenesis (28). This study revealed that FBXO24 regulates spatiotemporal mitochondrial dynamics during spermiogenesis. FBXO24 might serve as a critical protein in regulating sperm mitochondrial sheath formation, which expands the knowledge of sperm mitochondria formation. In addition, we found that FBXO24 is indispensable for proper flagellum formation, such as axoneme and ODFs. Therefore, the reduction of sperm motility of *Fbxo24* KO male mice observed in this study might attribute to the uncompleted mitochondrial formation of the middle piece and defective flagellum assembling.

It is worth mentioning that the RNA-seq data of the round spermatids showed that FBXO24 ablation leads to dysregulation in the mRNA expression of many genes related to chromatin, mitochondria, and flagellum. We found mRNA expression changes were related to the mRNA alternative splicing in *Fbxo24* KO round spermatids. Combined with our immunoprecipitation evidence of molecular interactions between FBXO24 and the key splice factors (SRSF2, SRSF3, and SRSF9), it is reasonable to infer that dysregulation of these splicing factors inevitably would lead to more splicing errors in their target genes, thus amplifying the initial adverse effects and generating a vicious circle of aberrant splicing. Of note, many genetic changes identified in this study have been closely associated with sperm formation. For example, *Zmynd12* (29), *Bbs7* (30), and *Ube2j2* (18) were required for flagellum function and male fertility. *Map7d1* facilitates microtubule stabilization (31) and *Mfn1* deficiency leads to defects in mitochondrial activity and male infertility (32). *Phf7* modulates BRDT stability and histone-to-protamine exchange during spermiogenesis (33). *Kat5* encodes an essential lysine acetyltransferase, which is involved in regulating gene expression and chromatin remodeling (34). *Nucb2* suppresses the acrosome reaction in sperm within the mouse epididymis (35). Therefore, our study, for the first time, provides evidence that FBXO24 interacts with the splicing factors to regulate the alternative splicing of mRNA involved in round spermatid development. However, the exact regulatory mechanism of how FBXO24 facilitates the mRNA alternative splicing needs further study.

MIWI and piRNAs are highly abundant in round spermatids, but their levels start to decline normally in elongating spermatids and are completely eliminated in matured sperm (36). Interestingly, Mo-Fang Liu et al. reported that piRNA-triggered MIWI degradation occurs in late spermatids through the APC/C-26S proteasome pathway in vitro, which in turn leads to piRNA elimination (37). As many piRNAs also appear to have the capacity to target diverse mRNA, they demonstrated that piRNAs act extensively as siRNAs to degrade specific mRNA (38, 39). Overexpression of a piRNA cluster in the mouse genome can reduce the expression of mRNA required for spermatogenesis, leading to male infertility (40). The level of 29–31nt piRNAs was remarkably increased in *Fbxo24* KO testes, which raises a possibility that piRNA-related transcriptional repression might contribute to the reduction of mature sperm in *Fbxo24* KO mice. In the current study, we identified that testis-enriched FBXO24 could interact with MIWI and mediate its degradation via the K48-linked ubiquitination, which provides new insight into MIWI degradation during spermiogenesis.

In summary, this study revealed a novel role for FBXO24 in controlling mRNA alternative splicing of round spermatids and MIWI/piRNA pathway, and demonstrated that FBXO24 is required for mitochondrial organization and chromatin condensation in mouse spermatozoa (Figure 10). The results disclosed FBXO24 interacts with splicing factors to regulate the mRNA splicing of some essential genes required for spermiogenesis, which probably functionally connects the dysregulation of genes involved in mitochondria migration and histone removal during spermiogenesis. FBXO24 interplays with the subunits of the SCF complex and mediated MIWI degradation. This study extensively expands our understanding of the mechanistic explanation of FBXO24 for the regulation of spermiogenesis.

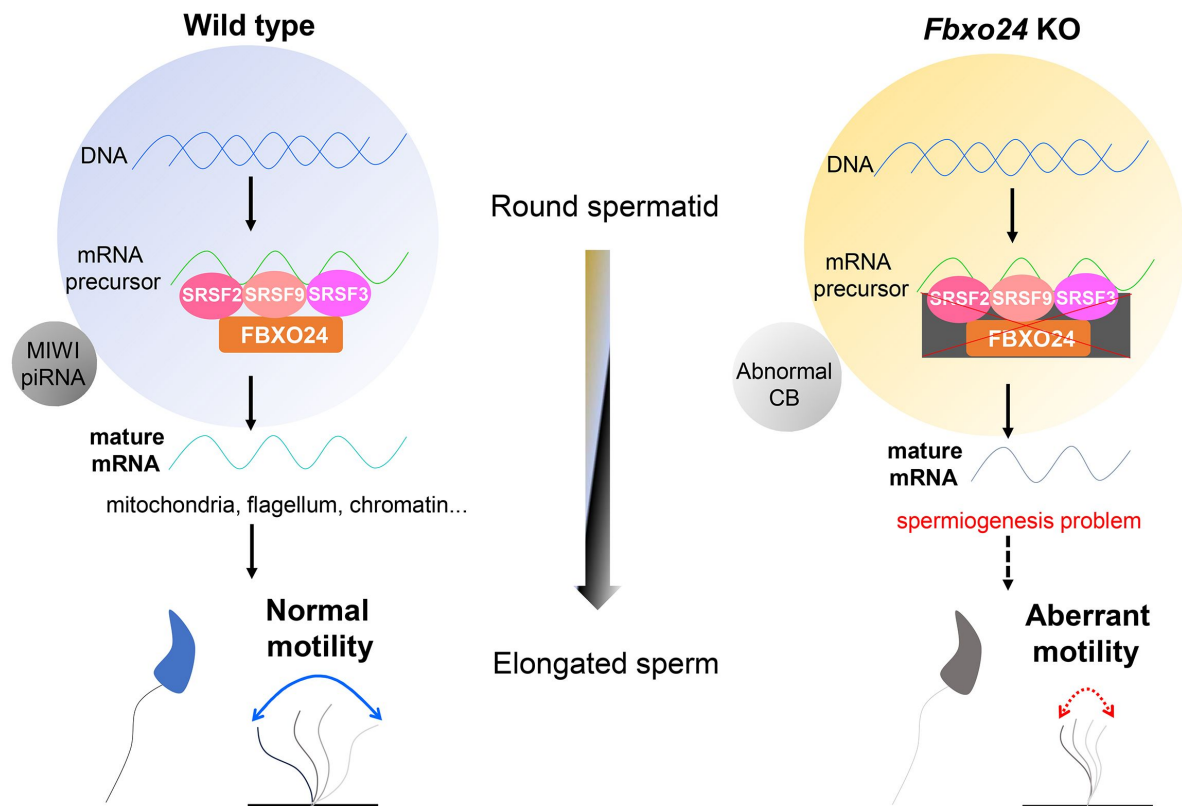


Figure 10.

A schematic model shows the FBXO24-mediated post-transcriptional regulation during spermiogenesis.

Fbxo24 interacts with key splicing factors (SRSF2, SRSF3, and SRSF9) to coordinate proper alternative splicing of the target mRNA transcripts involved in spermiogenesis. FBXO24 regulates the architectures of mitochondria and chromatin body through MIWI/piRNA pathway in the round spermatids.

Materials and Methods

Ethics statement

All the animal procedures were approved by the Institutional Animal Care and Use Committee of Tongji Medical College, Huazhong University of Science and Technology, and the mice were housed in the specific pathogen-free facility of Huazhong University of Science and Technology. All experiments with mice were conducted ethically according to the Guide for the Care and Use of Laboratory Animal guidelines.

Generation of *Fbxo24* mutant mice

Fbxo24 knockout (KO) mice were generated by CRISPR/Cas9 technology. Briefly, the two pairs of single-guided RNAs (sgRNAs) with the sequence sgRNA-1: 5'-TGTGGAGGCGCATCTGCGAAGG-3' and sgRNA-2 5'-GTCAAAGACTTGGTCGCCCTAGG-3' were designed for targeting for the third exon of *Fbxo24* gene. The sgRNAs were injected into the pronuclei of fertilized eggs, and the 2-cell stage embryos were transferred into the oviducts of pseudopregnant C57BL/6J females in the next day to generate *Fbxo24* mutant founder mice (F0). The F1 *Fbxo24* heterozygous mice were produced by crossing the F0 founder mice with WT mice and further intercrossed F1 to obtain *Fbxo24* KO mice carrying an 82-bp deletion in exon 3 of the *Fbxo24* gene. A 197-bp band as the wild-type (WT) allele and a 115-bp band as the deleted allele were designed for genotyping by PCR amplification. The primers used are listed in **Table S7**.

Generation of *Fbxo24*-HA tagged transgenic mice

Fbxo24-HA-tagged transgenic mice were also generated by CRISPR/Cas9 technology and haploid embryonic stem cell microinjection. The sgRNAs of the C-terminus of the target gene *Fbxo24* (AAGAAGAGGGGTTCAGCTCT) were synthesized, annealed, and ligated to the pX330-mCherry plasmid. For the construction of the HA-tag DNA donor, the sequences encoding the left homologous arm, the HA tag, and the right homologous arm were amplified and ligated to the linear pMD19T. DKO-AG-haESCs were transfected with CRISPR-Cas9 plasmid and HA tag DNA donor. At 24 h after transfection, the mCherry-positive haploid cells were enriched. In 7–8 days after plating, single colonies were picked up, and positive colonies for the HA tag were selected by genomic DNA PCR amplification for Sanger sequencing. DKO-AG-haESCs with *Fbxo24*-C-HA were arrested in M-phase by culturing in a medium containing 0.05 µg/ml demecolcine for 10 h and then used for intracytoplasmic injection as described previously (41). Intracytoplasmic AG-haESC injection (ICAHCI) embryos were cultured in KSOM medium for 24 h to reach the two-cell stage. 15–20 two-cell embryos were transferred into each oviduct of pseudo-pregnant females to produce *Fbxo24*-C-HA semi-cloned mice (F0). Then, the F0 female mice were crossed with WT male mice to generate heterozygous *Fbxo24*-C-HA F1 male mice for the experiments.

Antibodies

The details of all commercial antibodies used in this study are presented in **Table S8**.

Electron microscopy

For transmission electron microscopy (TEM), samples were fixed in 4% paraformaldehyde containing 0.05% glutaraldehyde in 0.1 M PBS, then post-fixed in 1% osmium tetroxide. Dehydration was carried out in ethanol (25, 50, 75, and 95%), and the samples were embedded in Epon 812. Ultrathin sections with 80-nm thickness were prepared by ultramicrotome (Leica UC7, Leica Biosystems, Germany). Ultrathin sections were counterstained with uranyl acetate and lead citrate and examined with a transmission electron microscope (Hitachi HT7700, Japan). For scanning electron microscopy (SEM), the samples were fixed in 2.5% glutaraldehyde solution in 0.1 M PBS and collected on poly-L-lysine-coated glass coverslips, followed by post-fixed in osmium

tetroxide and dehydrated in graded ethanol series. Then, the samples were subjected to critical point drying and coated with gold/palladium for observation with a scanning electron microscope (AZteclive Ultim Max 100, UK).

Sperm and histological analysis

The epididymis isolated from 8-week-old mice and sperm allowed to swim into M16 medium (Sigma-Aldrich), which was maintained at 37°C and 5% CO₂. After 30 min, the medium was collected and diluted for sperm number and motility assessment. To determine sperm percent motility, a 10 µL sample was loaded onto a clean slide glass and covered with a coverslip. Motility was graded according to the WHO criteria under positive phase contrast microscopy at a total magnification $\times 400$ (42 [42](#)). The length of midpiece was measured from the sperm neck to the sperm annulus by MitoTracker staining. All results were obtained from experiments performed on at least 6 mice per genotype group and 100 sperm per mouse. Testes and epididymis from WT and KO mice were fixed in Bouin's fixative and embedded in paraffin. For the histological analysis, sections of 5 µm were cut and stained with periodic acid-Schiff (PAS) or hematoxylin and eosin (H&E) after dewaxing and rehydration.

Immunofluorescence

Testes were fixed in 4% PFA in PBS overnight at 4°C and then were sequentially soaked in 5, 10, 12.5, 15, and 20% sucrose in PBS and embedded in Tissue-Tek O.C.T. compound (Sakura 4583, Sakura Finetek USA, Inc., Torrance, CA) on dry ice. 5 µm cryosections were cut and washed with PBS three times (10 min per wash), then the cryo-sections were microwaved in 0.01 M sodium citrate buffer (pH 6.0) and cooled down to room temperature (RT) for antigen retrieval. After washing with PBS three times, the sections were blocked in a blocking solution (containing 3% normal donkey serum and 3% fetal bovine serum in 1% bovine serum albumin) for 1 h. Then, the sections were incubated with primary antibodies overnight at 4°C and then incubated with secondary antibodies for 1 h at RT. After washing with PBS and stained with DAPI, the sections were photographed under FluoView 1000 microscope (Olympus, Japan) with a digital camera (MSX2, Micro-shot Technology Limited, China). Images were merged using Adobe Photoshop (Adobe Systems, San Jose, CA).

TUNEL staining

Testes were fixed in 4% PFA, embedded in Tissue-Tek O.C.T. compound, and cut section with 5 µm-thick. Sperm were fixed in 4% PFA and permeated by 0.3% Triton X-100. TUNEL staining was performed using the TUNEL ApoGreen Detection Kit (YEASEN, 40307ES20, China). Images were obtained with a FluoView 1000 microscope (Olympus, Japan).

Western blot

Fresh mouse adult testes or HEK293T cells were collected, and proteins were extracted by using RIPA buffer (Beyotime, P0013J, China). In total, 40 µg of protein lysates were separated on a 10% SDS-PAGE gel, proteins were transferred to PVDF membranes (Bio-Rad) and the membranes were blocked in 5% non-fat milk (blocking solution) for 1 h. Primary antibodies were incubated overnight at 4°C after blocking. The membranes were washed with TBST three times and then incubated with a secondary antibody for 1 h before using Luminol/enhancer solution and ClarityTM Western ECL Substrate (Bio-Rad Laboratories, Inc. US). Western blot images were scanned by using Gel Doc XR system (Bio-Rad Laboratories, Inc. US).

Immunoprecipitation (IP) assays

For testis tissue IP experiments, the testes were dissected and lysed in immunoprecipitation buffer (Beyotime, P0013J, China), clarified by centrifugation at 12,000 $\times g$, and then pre-cleared with anti-HA nanobody agarose beads (AlpaLife, ktsm1305, China). The lysate was incubated with primary

antibodies overnight at 4°C on a rotator and conjugated with anti-HA beads. The beads were washed with IP buffer and then boiled in 2× SDS loading buffer for western blotting analysis.

For cell line IP experiments, the HEK293T cells were transfected with indicated plasmids using Lipofectamine 2000 (Life Technologies). After 48 h, immunoprecipitation was performed as described previously (43). 25 µl mCherry-Trap bead 50% slurry (AlpaLife, ktsm1331, China) was used and all wash steps were performed with washing buffer (10mM Tris-HCl pH7.5, 150mM NaCl, 0.5mM EDTA). mCherry-Trap beads were washed with dilution buffer prior to addition to the cell lysate. Beads were incubated with cell lysate at 4°C for 2 h following another wash step. To elute the proteins from the beads, 40 µl sample buffer (120 mM Tris-HCl pH=6.8, 20% glycerol, 4% SDS, 0.04% Bromophenol blue, 10% β-mercaptoethanol) was added and boiled at 95°C for 5 min. mCherry-tagged proteins were detected by western blotting using an anti-mCherry antibody.

Spermatogenic cell isolation

Spermatogenic cells were isolated from the whole mouse testis using the STA-PUT method described previously with slight modifications (44). In brief, testes were collected from 8-week-old mice WT and *Fbxo24* KO mice and decapsulated with collagenase treatment to remove Leydig cells. The dispersed seminiferous tubules were then digested with trypsin and DNase I to single-cell suspensions. Next, Sertoli cells were separated from germ cells by filtration through a 40 mm cell strainer and by adhesion to lectin-coated culture plates. Store the cells in the 1× Krebs buffer on ice. Different germ cell populations were separated using a manually prepared 0.5–5% discontinuous BSA density gradient for velocity sedimentation sediment. After sedimentation (1.5 h, 4°C), enriched fractions of the cell were manually collected, and the cell concentration was determined.

RNA isolation and quantitative qPCR

Total RNAs were extracted from purified germ cell fractions using TRIzol reagent (Invitrogen, UK) following the manufacturer's procedure. The purity and concentration of RNA samples were determined using a Nanodrop ND-2000 spectrophotometer (Thermo Scientific, Madison, USA). Reverse transcription of 500 ng purified total RNA was performed by PrimeScript RT reagent kit with gDNA Eraser (Takara, Dalian, China). RT-qPCR was performed with SYBR green master mix (Takara, Dalian, China) on the ABI Step One System (Applied Biosystems) according to manufacturers' procedure. The relative gene expression was quantified using the comparative cycle threshold method, with the *Gapdh* expression used for normalization.

RNA-seq analysis

1 µg of total RNA was used from each group to prepare the mRNA libraries using TruSeq Stranded mRNA Library Preparation Kit Set A (Cat. No. RS-122-2101, Illumina) according to the manufacturer's instructions. All libraries were sequenced using the Illumina HiSeq 4000 platform. The FASTX-Toolkit was used to remove adaptor sequences, and low-quality reads from the sequencing data. To identify all the transcripts, we used Tophat2 and Cufflinks to assemble the sequencing reads based on the UCSC mm10 mouse genome. The differentially expression analysis was performed by Cuffdiff. The differential expressed genes were set with the threshold of $P < 0.05$ and fold-change > 2 .

Small RNA-seq and annotation

For small RNA-seq, the total RNA from adult testes was gel fractionated, and those 16–40 nt in length were enriched by polyacrylamide gel electrophoresis, and purified for small RNA-seq. The small RNA libraries were constructed using the Digital Gene Expression for Small RNA Sample prep kit (Illumina). Sequencing of the small RNA library was performed by Illumina HiSeq Xten. Small RNA annotation was performed as described previously (45).

Statistical analysis

All data are presented as mean $\pm$ S.D. unless otherwise noted in the figure legends. Statistical differences between datasets were assessed by one-way ANOVA or Student's *t*-test using the GraphPad Prism 8 software. *P*-values are denoted in figures by **p* < 0.05; ***p* < 0.01; ****p* < 0.001.

Data availability

All data needed to evaluate the conclusions in the paper are present in the article and/or the Supplementary Materials. RNA-seq and Small RNA-seq data are deposited in the NCBI SRA database with the accession number PRJNA878933 and PRJNA878953, respectively.

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Author contributions

Z.L., L.Z., and S.Y. conceived and designed the research. Z.L., X.L., Y.Z., and Y.L. performed the bench experiments and data analysis. L.Z. generated the Fbxo24 KO mouse model. Z.L. wrote the manuscript. S.Y., L.Z., and Z.L. supervised the project. S.Y., and Z.L. revised the manuscript. All authors read and approved the manuscript.

Conflict of interest

The authors declare no conflict of interest.

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Reviewer #1 (Public Review):

In this study, Li et al., report that FBXO24 contributes to sperm development by modulating alternative mRNA splicing and MIWI degradation during spermiogenesis. The authors demonstrated that FBXO24 deficiency impairs sperm head formation, midpiece compartmentalization, and axonemal/peri-axonemal organization in mature sperm, which causes sperm motility defects and male infertility. In addition, FBXO24 interacts with various mRNA splicing factors, which causes altered splicing events in *Fbxo24*-null round spermatids. Interestingly, FBXO24 also modulates MIWI levels via its polyubiquitination in round spermatids. Thus, the authors address that FBXO24 modulates global mRNA levels by regulating piRNA-mediated MIWI function and splicing events in testicular haploid germ cells.

This study is performed with various experimental approaches to explore and elucidate underlying molecular mechanisms for the FBXO24-mediated sperm defects during germ cell development. Overall, the experiments were designed properly and performed well to support the authors' observation in each part. In addition, the findings in this study are useful for understanding the physiological and developmental significance of FBXO24 in the male germ line, which can provide insight into impaired sperm development and male infertility.

In the revised manuscript, the authors address most of the concerns raised in the previous review. The following are representative remaining points.

- Quantification of the defective, vacuolar mitochondria (80%) and missing annulus (30%) was not shown in the figures or described in the results as well as in a few other figures.

<https://doi.org/10.7554/eLife.91666.2.sa2>

Reviewer #2 (Public Review):

Spermatogenesis describes a complex sequence of differentiation events that lead to the development of genetically distinct male germ cells. The final part of spermatogenesis is called spermiogenesis, in which spermatids differentiate into mature sperm by developing an acrosome and a motile flagellum, which are required for reaching and successfully penetrating the oocyte. This process of spermatogenesis is based on a coordinated regulation of gene expressions in round spermatids. In the current study, FBXO24 was identified as a highly expressed protein in human and mouse testis. To define its biological role *in vivo*, the authors generated genetically engineered *Fbxo24* knockout and *Fbxo24*-HA-labeled transgenic mouse models.

To elucidate the causes of the observed sterility in *Fbxo24*-KO males, the authors performed molecular and phenotypic analyses that revealed aberrant histone retention, incomplete axonemes, oversized chromatoid bodies (CB), and abnormal mitochondrial coiling along the sperm flagella. These results support the causal role of the FBXO24 gene in sperm motility.

Furthermore, the authors carefully characterized by SEM, TEM and western blot analyses that deletion of FBXO24 leads to incomplete histone-to-protamine exchange and defective chromatin interaction during spermiogenesis. In addition to increased MIWI expression, the authors show that FBXO24 interacts with SCF subunits and mediates the degradation of MIWI via K48-linked polyubiquitination.

This is a solid work demonstrating the role of FBXO24 in modulating alternative mRNA splicing, MIWI degradation and normal spermiogenesis.

Reviewer #3 (Public Review):

This work is carried out by the research group led by Shuiqiao Yuan, who has a long interest and significant contribution in the field of male germ cell development. The authors study a protein for which limited information existed prior to this work, a component of the E3 ubiquitin ligase complex, FBXO24. The authors generated the first FBXO24 KO mouse model reported in the literature using CRISPR, which they complement with HA-tagged FBXO24 transgenic model in the KO background. The authors begin their study with a very careful examination of the expression pattern of the FBXO24 gene at the level of mRNA and the HA-tagged transgene, and they provide conclusive evidence that the protein is expressed exclusively in the mouse testis and specifically in post-meiotic spermatids of stages VI to IX, which include early stages of spermatid elongation and nuclear condensation. The authors report a fully sterile phenotype for male mice, while female mice are normal. Interestingly, the testis size and the populations of spermatogenic cells in the KO mutant mice show small (but significant) reduction compared to the WT testis. Importantly, the mature sperm from KO animals show a series of defects that were very thoroughly documented in this work by scanning and transmission electron microscopy; this data constitutes a very strong point in this paper. FBXO24 KO sperm have severe defects in the mitochondrial sheath with missing mitochondria near the annulus, and missing outer dense fibers. Collectively these defects cause abnormal bending of the flagellum and severely reduced sperm motility. Moreover, defects in nuclear condensation are observed with faint nuclear staining of elongating and elongated spermatids, and reduction of protein levels of protamine 2 combined with increased levels of histones and transition protein 1. All the above are in line with the observed male sterility phenotype.

The authors also performed RNASeq in the KO animal, and found profound changes in the abundance of thousands of mRNAs; changes in mRNA splicing patterns were noted as well. This data reveals deeply affected gene expression patterns in the FBXO24 KO testis, which further supports the essential role that this factor serves in spermiogenesis. Unfortunately, a molecular explanation of what causes these changes is missing; it is still possible that they are an indirect consequence of the absence of FBXO24 and not directly caused by it.

The finding that Miwi protein levels are increased in the FBXO24 KO testis is an important point in this work, and it is in agreement with the observed increased size of the chromatoid body, where most of Miwi protein is accumulated in round spermatids. This finding is well supported with experiments performed in 293T cells showing that Miwi ubiquitination is FBXO24 dependent in this ectopic system. Moreover, the authors detect reduced ubiquitination of endogenous Miwi protein immunoprecipitated from FBXO24 KO testis. Consistent with an increase in Miwi protein levels, Miwi-sized piRNAs show increased abundance in total RNA from FBXO24 KO testis. It has been documented that Piwi proteins stabilize their piRNA cargo, so the increase in piRNA levels in 29-32 nt sizes is most likely not a result of altered biogenesis, but increased half-life of the piRNAs as a result of Miwi upregulation. piRNAs have been involved in the regulation of mRNAs in the post-meiotic spermatid, but it is unclear how increased Miwi protein and its piRNA cargo at the levels observed in this study contribute to the complete infertility phenotype of the FBXO24 KO male mice.

Therefore, a well-reasoned narrative on if and how the absence of FBXO24 as an E3 ubiquitin ligase is responsible for the observed mRNA and protein differential expression is largely absent. If FBXO24-mediated ubiquitination is required for normal protein degradation during spermiogenesis, protein level increase should be the direct consequence of genuine FBXO24 targets in the KO testis. Apart from Miwi, the possible involvement of ubiquitination was not

shown for any other proteins that the authors found interact with FBXO24 such as splicing factors SRSF2, SRSF3, SRSF9, or any of the other proteins whose levels were found to be changed (reduced, thus the change in the KO is less likely due to absence of ubiquitination) such as ODF2, AKAP3, TSSK4, PHF7, TSSK6 and RNF8. Interestingly, the authors do observe increased amounts of histones and transition proteins, but reduced amounts of protamines, which directly shows that histone to protamine transition is indeed affected in the FBXO24 KO testis, consistent with the observed less condensed nuclei of spermatozoa. Could histones and transition proteins be targets of the proposed ubiquitin ligase activity of FBXO24, and in its absence, histone replacement is abrogated? Providing experimental evidence to address this possibility would greatly expand our understanding on why FBXO24 is essential during spermiogenesis.

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Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

However, there are several concerns to be explained more in this study. In addition, some results should be revised and updated.

Thank you for your comments. The concerns were addressed by the description and experiment.

Some results were revised and updated accordingly.

Reviewer #2 (Public Review):

The minor weakness of the study is inconsistent use of terminology throughout the manuscript, occasional logic-jump in their flow, and missing detailed description in methodologies used either in the text or Materials and Methods section, which can be easily rectified.

Thank you for your review. We have revised the manuscript and corrected errors according to your comments.

Reviewer #3 (Public Review):

Importantly, besides the Miwi ubiquitination experiment which is performed in a heterologous and therefore may not be ideal for extracting conclusions, the possible involvement of ubiquitination was not shown for any other proteins that the authors found that interact with FBXO24. Could histones and transition proteins be targets of the proposed ubiquitin ligase activity of FBXO24, and in its absence, histone replacement is abrogated?

Thank you for your comments. The histones and transition proteins were not found in the immunoprecipitates of FBXO24, suggesting they are not the direct targets of FBXO24, shown in Figure S3G.

Miwi should be immunoprecipitated and Miwi ubiquitination should be detected (with WB or mass spec) in WT testis.

We agree with this suggestion. In the revision, the expression and ubiquitination of MIWI were detected in WT testis by the immunoprecipitation and ubiquitination assay, as shown in Figure 8H.

Therefore, the claim that FBXO24 is essential for piRNA biogenesis/production (lines 308, 314) is not appropriately supported.

We appreciate the comment. We have revised the description and modified the claim on page 11.

Reviewing Editor's note for revision

(1) As noted by all three reviewers, as currently written the rationale to focus on MIWI is not entirely clear. A transitional narrative to focus on MIWI needs to be provided as well as an explanation for how the absence of FBXO24 as an E3 ubiquitin ligase is responsible for the observed mRNA and protein differential expression.

We appreciate your comments. We have supplemented the transitional narrative by focusing on MIWI and explained mRNA and protein differential expression upon FBXO24 deletion, shown on Page 7 and Page 13, respectively.

(2) As it can be indirect, mass spec detection of MIWI in testis co-IP and MIWI ubiquitination should be detected (with WB or mass spec) in WT testis.

In the revision, the expression and ubiquitination of MIWI were detected in WT testis by the immunoprecipitation and ubiquitination assay, as shown in Figure 8H.

(3) Please tone down the claim that FBXO24 is essential for piRNA biogenesis/production as it requires further evidence.

We have revised the description and modified the claim on page 11.

(4) Ontology analysis of the genes with abnormally spliced mRNAs to provide an explanation for developmental defects.

In the revision, we have performed the ontology analysis and provided new data regarding the abnormally spliced genes, as shown in Figure S4D.

Reviewer #1 (Recommendations For The Authors):

Major comments

(1) The authors performed mainly with the WT (or knock-in) and Fbxo24-knockout mouse model. Do the heterozygous males and their sperm have any physiological defects like FBXO24-deficient mice?

This is a good question. We did the phenotype analysis and found that heterozygous males are all fertile, and their sperm do not have any physiological defects.

(2) Fbxo24-KO sperm carries swollen mitochondria. How do the mitochondria affect sperm function?

Thank you for raising this interesting question. Based on our data and published literature, the defective mitochondria were associated with energetic disturbances and reduced sperm

motility, as shown on Page 12.

(3) TEM images show that Fbxo24-KO spermatids carry swollen mitochondria and enlarged chromatoid bodies. How the swollen mitochondria and enlarged chromatid are defective for sperm motility and flagellar development, requires more explanation. In addition, it is unclear how the enlarged diameter of the chromatoid body is critical for normal sperm development.

Thank you for your comments. The chromatoid bodies are considered to be engaged in mitochondrial sheath morphogenesis. Analysis of the chromatoid bodies' RNA content reveals enrichment of PIWI-interacting RNAs (piRNAs), further emphasizing the role of the chromatoid bodies in post-transcriptional regulation of spermatogenetic genes. We added this explanation on Page 12-13.

(4) The authors only show band images to compare the protein amounts between WT and KO sperm and round spermatids. As the blots for loading controls are not clear, the authors should quantify the protein levels and perform a statistical comparison.

We quantified the protein levels and performed a statistical comparison, as shown in Figure S3B.

(5) The authors show the defective sperm head structure from Fbxo24-KO sperm in Figure 5. However, the Fbxo24-KO sperm heads seem quite normal in Figure 3. How many sperm show defective sperm head structure? In addition, the authors observed altered histone-to-protamine conversion in sperm, but it is unclear whether the altered nuclear protein conversion causes morphological defects in the sperm head.

We appreciate the comments. In our study, we found over 80% of Fbxo24 KO sperm showed defective structure in the sperm head. Altered histone-to-protamine conversion caused the decondensed nucleus of Fbxo24 KO sperm. Notably, in many knockout mice studies, impaired chromatin condensation is frequently associated with abnormal sperm head morphology, as shown in reference 15 of Page 8.

(6) The authors compare the protein levels of RNF8, PHF7, TSSK6, which participate in nuclear protein replacement in sperm. However, considering the sperm is the endpoint for the nuclear protein conversion, it is unclear to compare the protein levels in mature sperm. The authors might want to compare the protein levels in developing germ cells.

Thank you for your comment. Yes, we actually detected the protein levels of RNF8, PHF7, and TSSK6 in the testes, not in sperm. We have corrected it in the Figure 5E. We apologize for our carelessness.

(7) This reviewer suggests describing more rationales for how the authors focus on the MIWI protein. Also, it is wondered whether MIWI is also detected from testis co-IP mass spectrometry.

We agree with this suggestion. Since MIWI was a core component of CB and also identified as an FBOX24 interacting partner from our immunoprecipitation-mass spectrometry (IP-MS) (Table S1), we focused on the examination of MIWI expression between WT and Fbxo24 KO testes. We have added this description in the revision (see lines 191-193 on page 7).

(8) The authors need to provide a more detailed explanation for how the altered piRNA production affects physiological defects in germ cell development. In addition, it will be good to describe more how the piRNAs affect a broad range of mRNA levels.

Thank you for your comments. The previously published studies have demonstrated that piRNAs could act as siRNAs to degrade specific mRNAs during male germ cell development and maturation. We have cited these studies on lines 369-372 of Page 13.

(9) The authors observed an altered splicing process in the absence of FBXO24. However, it is a little bit confusing how the altered splicing events affect developmental defects. Therefore, the authors should state which mRNAs have undergone abnormal splicing processes and provide ontology analysis for the genes.

We have performed the ontology analysis and showed the new data in Figure S4D.

Minor comments

(1) Figure 1A-C - Statistical comparison is missed. Numbers for biological replication should be described in corresponding legends.

Thank you for your careful review. We have provided the statistical comparison and the numbers for biological replication in the legends of Figure 1A-C.

(2) Figure 1E, F - Current images can't clearly resolve the nuclear localization of the FBXO24 testicular germ cells. To clarify the intracellular localization, the authors should provide images with higher resolution.

The resolution of Figure 1E, F was improved, as suggested. Thank you!

(3) Figure 1E, F - Scale bar information is missing.

The scale bars of Figure 1E, F were provided.

(4) It will be much better to show the predicted frameshift and early termination of the protein translation in Fbxo24-knockout mice.

The predicted frameshift of Fbxo24-knockout mice was added and shown in Figure S1B.

(5) It is required to provide primer information for qPCR.

The primer information for qPCR was provided, as shown in Table S7.

(6) The authors describe that Fbxo24-KO sperm show abrupt bending of the tail. However, the description is unclear and the sperm shown in Figure 3C seems quite normal. The authors should clarify the abnormal bending pattern of the tail and show quantified results.

Thank you for pointing out this issue. In Fbxo24 KO sperm, abnormal bending of the sperm tails mainly included neck bending and midpiece bending. We have shown them in Figure S3A.

(7) The authors mention that Fbxo24-KO sperm have swollen mitochondria at the midpiece, but this is also unclear. How many mitochondria are swollen in Fbxo24-KO sperm?

This is a good question. However, since it is very difficult to observe all of the mitochondria in each sperm using the electronic microscope, we could not quantify the swollen mitochondria in Fbxo24 KO sperm.

(8) Scale bar information is missed - Fig 3C insets, Fig 3D, Fig 3F insets, 4A insets, Figure 4C insets.

All the scale bars have been added.

(9) How many sperm have annulus defects? In Figure 3F, WT sperm does not have an annulus, which could be damaged during sample preparation. Is the annulus defects in Fbxo24-KO sperm consistent?

Thank you for asking these questions. Based on our results, about 30% of Fbxo24 KO sperm showed defective annulus structure. Since both TEM (Figure 3F) and SEM (Figure 3G) results clearly showed the defective annulus structure of Fbxo24 KO sperm, we believe the annulus defects are consistent and highly unlikely caused by sample preparation.

(10) A Cross-section image for the endpiece of Fbxo24-KO sperm is not suitable. There is a longitudinal column structure of the principal piece.

Thank you for your comments. It is difficult to observe a completely longitudinal structure of sperm tail under TEM. The cross-section of the endpiece and principal piece allowed us know the structure of the axoneme, ODFs and fibrous sheath (FS).

(11) The endpiece of Fbxo24-KO sperm seems to have a normal axoneme. Do all endpieces of Fbxo24KO sperm have normal axoneme? Also, the authors need to describe whether an axonemal structure is damaged and disrupted in all Fbxo24-KO sperm.

Our TEM data showed the axonemal structure was impaired in the endpiece of Fbxo24 KO sperm (See right panels of Figure 3H). Moreover, based on the ultrastructure analysis of TEM, we found over 90% of Fbxo24 sperm had a damaged axonemal structure.

(12) Reference blots in Fig 3I, 3J, 4E (left), 5C and 5E are quite faint. The authors should replace the blot images.

Thank you for pointing out this. We have rerun Western blot multiple times but could not obtain better images due to antibody sensitivity. However, we quantified the protein levels and performed a statistical comparison, as shown in Figure S3B, to establish a good readout from these images for the readers.

(13) Loading controls are required - 7D-H.

Done as suggested. Thanks!

(14) How do the authors measure the midpiece length? From where to where? This should be clarified.

Good question. We measured the midpiece length from the sperm neck to the sperm annulus by MitoTracker staining. We have clarified this on Page 16.

(15) How are the bands for Fbxo24 shifted during IP in Fig 7A?

The protein modification in the interaction may cause the band shift.

(16) There are several typos throughout the manuscript. Please check carefully and fix them.

Thank you for your careful review. We have corrected and fixed all the typos as far as we can.

Reviewer #2 (Recommendations For The Authors):

Major comments

(1) Please provide a schematic of HA-Fbxo24 knock-in construct and strategy together with knockout (Figure S2) or even separately early in Figure S1. The description of using the transgenic mouse is mentioned even earlier than the knockout but there are no citations or methods provided in the text other than that listed in Materials and Methods.

Thank you for your suggestion. As suggested, the schematic of the HA-Fbxo24 knock-in strategy has been supplemented in Figure S2A. The description of using the transgenic mouse has been added to the results, as shown on page 4 of lines 102-103.

Also, it is not clear to what extent the phenotypic and molecular characterization of HA-transgenic mice is performed. For example, Lines 134-139: The use of Fbxo24-HA labeled transgenic mice results in the rescue of spermatogenesis and fertility as shown in Figure 2F by measuring the litter size. It is not clear how this observation leads the author to state that this rescues defects in spermiogenesis. Please clarify how and what other measures are taken to support this conclusion. Is the observed infertility due to defects in spermatogenesis or spermiogenesis?

Thank you for your question. We crossed FBXO24-HATag males with FBXO24^{-/-} females to obtain FBXO24^{-/-}; FBXO24-HATag males. We examined the testes volume and histological morphology of FBXO24^{-/-}; FBXO24-HATag males and found that they were similar to FBXO24^{+/-}; FBXO24-HATag littermates, indicating that spermatogenesis was restored, as shown in Figure S2H.

(2) Line 107 vs Line 114: Please use the terminology spermatogenesis and spermiogenesis consistently throughout the text. Earlier in the introduction, the authors clearly defined that spermatogenesis involves three phases, with the third phase referred to as spermiogenesis. However, the author concludes in the first line that "FBXO24 plays a role during spermatogenesis" while summarizing at the end of the paragraph that this protein is "expressed in haploid spermatids specifically during spermiogenesis". Therefore, it is not clear whether the authors conclude that FBXO24 is important for all of spermatogenesis (line 107) or only for part of spermiogenesis (line 114). Another example is line 219 vs. 238: At this point in the manuscript, it is again unclear whether the authors want to study molecular changes during spermatogenesis or spermiogenesis upon FBXO24 depletion. Many examples of such cases throughout the text, and it is recommended to be consistent in using more restrictive terminology whenever applicable for a clear interpretation.

We thank you for your careful review. We have double-checked the terminology of spermatogenesis and spermiogenesis and made it consistent throughout the text of the revised manuscript.

(3) It is not clear how rampant/frequent the Fbxo24-knockout sperm show defects in head morphology based on Figures 3C, 3F, and 5A since it seems that there are some sperm showing relatively normal looking sperm heads. Please provide quantification.

We have performed the quantification and found that over 80% of Fbxo24 KO sperm showed defective structures in the sperm head.

(4) Figure 3B: The authors describe in the figure legend that 3 mice were analyzed in each group. The standard deviation for the WT analysis is missing, or if the author wanted to set the WT value to 100%, the bar and scale shown on the y-axis do not fit. The value for WT looks more like 95%.

We have indeed analyzed sperm motility based on the WT value set at 100% and have revised Figure 3B in the revision. We apologize for this oversight.

(5) Figure 3 B and C: It is not clear how the motility is measured. Is CASA used (not described in Methods). The conclusion about abnormal flagellar bending in KO spermatozoa cannot be drawn from the static microscopic images alone. Please provide more details of motility analysis together with videos of live cell imaging.

The sperm motility was measured manually using a hemocytometer, according to the reference.

We provided the details of sperm motility analysis in the Materials and Methods section on Page 16.

(6) Figure 3 I and J: These are one of a few figures that are not supported by statistical analysis. In particular, for 3I, GAPDH controls of WT and KO protein do not show equal loading, which could explain the lower expression of the KO protein. Please show normalized bar graphs with multiple biological replicates or at least show a representative technical replicat that shows equal loading of GAPDH to better support the conclusion.

Thank you for your suggestion. Statistical comparison of relative protein expression was supplemented, as shown in new Figure S3B.

(7) Line 184: It is not clear how the authors define a swollen mitochondrion? Are there any size criteria (roundness) that can be measured to distinguish between a swollen and a non-swollen mitochondrion? It is recommended to use another terminology as often 'swollen' implies there is a difference in osmolarity but there is no experiment to support this implication.

Thank you for your comment. We have changed the “swollen” to “vacuolar” in the revision, as shown on Page 7.

(8) Figure S4, without a bright field image, it is hard to see the purity and morphology of the isolated prep. Please provide the bright field images together or as overlaid images.

We agree with your comment. We have provided the overlaid images in new Figure S4A.

(9) There is a big logic jump in what prompts the authors to look MIWI protein level and link the observation to MIWI/piRNA pathway in both Introduction and Results while it is one of the main findings. It is recommended to provide a better rationale and logical flow in the text.

Thank you for your suggestion. We have added a sentence explaining why we wanted to focus on studying MIWI expression (see lines 190-193 on page 7).

Minor comments

(1) Please keep all the conventions of gene vs. protein nomenclature. For example, write the genes mentioned in the figures in italics with the first letter in Capital, as it is done in

the main part. Proteins should be in ALL CAPITAL like FBXO24.

The names of gene and protein have been revised in the revision, as suggested.

(2) In the MM section, the name of the manufacturer and the location of the materials used are missing in several sections. Please go back through the MM section and add this information in the appropriate places.

Done as suggested. Thank you!

(3) On page 4, the authors mentioned that "Further qPCR analysis of developmental testes and purified testicular cells showed that FBXO24 mRNA was highly expressed in the round spermatids and elongating spermatids (Fig 1B-C)". Please include statistical analyses for Fig 1B-C as well as for Fig 1A to support the written statements.

Statistical comparison was supplemented, as shown in Figure 1. P-values are denoted in figures by * $p < 0.05$.

(4) Figure 3E: Please describe in more detail how the length of the midpiece was measured. Was it based on TEM images or based on fluorescent images using MitoTracker?

As we responded to Reviewer #1, we measured the midpiece length from the sperm neck to the sperm annulus by MitoTracker staining. We have clarified this in the Method and Material section on Page 16.

(5) Line 431: In the "Electron Microscopy" section of the MM part, the author should indicate the ascending ethanol series (%) used.

Done as suggested. Thank you!

(6) Line 432: The thickness of the sections prepared is missing, as well as an indication of the microtome used.

We have added thickness and the microtome in the Method and Material section on Page 16.

(7) Line 433: If the generated tiff files have been processed with Adobe Photoshop, this information is missing.

We have provided information on the usage of Adobe Photoshop for the generation of tiff files on Page 17.

(8) Lines 445, 452, 467: In some places in the paper, the temperature is written with a space between the number and {degree sign}C, and sometimes it is not. Please go through the paper and make it consistent. The usual spelling is 4{degree sign}C.

We have gone through the manuscript and checked all the spelling of temperature writing to make them consistent. Thank you for careful review.

(9) Line 469: The gel documentation system used is not mentioned.

Done as suggested. Thank you!

(10) Line 469: The 'TM' should be superscripted.

Done as suggested.

| (11) Line 489: A space is missing between the changes and the parenthesis.

Done as suggested.

| (12) Line 495-496: The authors write that the fractions enriched with round spermatids after sedimentation were collected manually. Was a determination of cell concentration - e.g., 2×10^6 cells/ml - performed after collection of the cells? How were the cells stored until use? Please add the sedimentation time and used temperature.

Store the cell in the 1 Krebs buffer on ice. The cell sediment was through a BSA density gradient for 1.5 h at 4°C. The cell concentration was determined after collection, as shown on Page 18.

| (13) Line 505: spelling error. Instead of "manufacturer's procedure" it is written manufactures' instructions.

The spelling error was corrected.

| (14) Line 520: Please write a short sentence on how the purification of the 16-40 nt long RNA was performed.

The length of 16–40 nt RNA was enriched by polyacrylamide gel electrophoresis. We added this information on Page 19 of line 531.

| (15) Line 528: The version of the used GraphPad software is missing.

The version of GraphPad software was supplemented, as shown on Page 19.

| (16) Line 677: For qPCR analyses, the number of mice analyzed (N) and a statistical evaluation are missing.

The statistical comparison and the numbers for biological replication were added, as shown on Page 26.

| (17) Figure 3D: Please add a scale bar.

Done as suggested. Thanks!

| (18) Line 371 and Line 377: Two times "in summary" is written. Please make one summary for the whole paper.

This sentence was revised, as shown in Page 13.

| (19) Line 382: To be consistent in the whole paper, please write Figure 10 in bold letters.

Done as suggested.

| (20) Please make the size and font of the references consistent with the main text.

Done as suggested. Thanks again for your careful review.

Reviewer #3 (Recommendations For The Authors):

I would like to see the description of the FBXO24 immunoprecipitation experiment performed in HEK293T cells. This somatic cell line does not normally express Miwi, so how Miwi was detected in FBXO24 mCherry IP beads? It is not mentioned if Miwi is expressed from a recombinant vector in this experiment. Similarly, I would like to see a better description of the experiment described in the same paragraph towards the end of it with the ubiquitin peptides, it is not clear.

Thank you for your comments. FBXO24-mCherry was expressed in HEK293T cells and the immunoprecipitates was incubated with the protein lysate of the testes (see lines 268-272 on Page 10). The description of the ubiquitin experiment was added as well, as shown in lines 283-286 on Page 10.

Line 263: I think the term ectopic here is not appropriate, a correction is needed.

We have changed “ectopic” to “increased” in the revision (see line 268 on Page 10).

I would like the authors to provide a tentative explanation or evidence of why FBXO24 KO males are completely sterile, even though there are still mature sperm produced with some motility. Since there are defects in nuclear condensation it will be very relevant to check DNA damage/fragmentation, which could contribute to the sterility phenotype.

This is a good suggestion. We reanalyzed the sperm DNA damage by TUNEL staining and shown the new data in Figure S3E-F.

Line 213: There have been some conflicting reports about the role of RNF8 in spermiogenesis, but a recent report has shown that RNF8 is not involved in histone PTMs that mediate histone to protamine transition (Abe et al Biol Reprod 2021 <https://doi.org/10.1093%2Fbiolre%2Fioab132>).

Thank you for your comment. We have cited this critical reference and discussed it in Discussion section on Page 12.

Figure 7: I would like to see zoomed-out views of the affected exons, so that flanking unaffected exons can be used as a reference for unaffected splicing. Most of the genome browser views in this image only show affected exons and it is impossible to see if these alone are affected or if the reduced RNAseq coverage in those exons is a result of overall reduced mapped reads in these genes. Also, a fixed Y axis with the same max value should be shown for these genome browser snapshots so that the expression level is comparable between the two genotypes.

Thank you for your comments. Loading control of RT-PCR and scale range of Y axis were added in new Figure 7.

Minor corrections:

Line 70: correct “..functions as protein-protein interaction..”.

Thank you for your careful review. We have corrected this sentence (see line 69 on Page 3).

Line 101: correct “..qPCR analysis of developmental testis..”.

We have corrected this sentence (see line 100 on Page 4). Thanks again.

| *Line 116: correct "..results in detective..".*

Corrected.

| *Line 186: correct ".. explored..".*

Corrected.

| *Line 218: correct ".. gene expressions.*

Corrected.

| *Line 221: correct "..genes significantly differentiated expressed".*

Corrected.

| *Line 241: FBXO24 was shown earlier in both cytoplasm and nucleus.*

We have changed "FBXO24 is mainly confined to the nucleus" to "FBXO24 expressed in the nucleus", as shown in line 247 on Page 9.

| *Line 501-502: correct "..reverse transcriptional".*

"reverse transcriptional" was changed into "reverse transcription", showing in Page 18.

| *Line 686: correct ".. deficiency male..".*

Corrected.

| *Line 769: correct "..Western blots were adopted..".*

Corrected.

| *Line 784: correct "..WT tesis..".*

Corrected.

| *I cannot understand exactly what is shown in Figure 9B. Some elements marked on the X-axis are single base locations (-2K, TSS, +2K) and others are stretches of sequences so they cannot be equivalent. Why there is only an intron shown? There should be a measure of normalized expression on the Y-axis.*

Thank you for your questions. The X-axis means that genome segments were scaled to the same size and were calculated the signal abundance, which was analyzed by computeMatrix. Aim to know the piRNA source, piRNA was mapped to the gene body, including introns, CDS and UTRs. The value of the Y-axis is the normalized count.

| *Figure 6F is not needed.*

Figure 6F was used to illustrate the number of different types of mRNA splicing upon FBXO24 deletion in the round spermatids. To better understand the splicing for the reader, we decided to keep it.

| *The last two paragraphs of the discussion seem to be redundant.*

Thank you for pointing out this. We have revised the last two paragraphs of the discussion.